Highlights

Effect Identity for LLM Agents:
Schema-Determined Limits on Duplicate Suppression

Sivaguru R. M.

- Retrying LLM agents paraphrase irreversible calls, so content hashes miss duplicates.
- A schema-fixed certifier and refuter leave only a residual for an LLM judge.
- A conformal rule on the max over m candidates bounds silent wrong suppression by α.
- The bound needs a jurisdiction-matched pool; a wider pool hides its own violation.
- Refining the schema shrinks the calibration budget: an exact refinement dilemma.

Effect Identity for LLM Agents: Schema-Determined Limits on Duplicate Suppression

Sivaguru R. M.[a,*]

[a]*Independent Research, AgentRuntime Project, Chennai, Tamil Nadu, India*

Abstract

An autonomous agent that retries an irreversible action re-synthesises its arguments, so two requests for the same real-world act may share no surface token. Content-addressed keys then miss the duplicate, and an unconstrained language-model judgement offers no bound on the opposite, silent error of suppressing a distinct act. We treat effect identity as a decision under asymmetric loss with a stochastic oracle and adversarial input. Any identity procedure decomposes into a certifier that proves sameness, a refuter that proves difference, and a residual that neither settles, determined by the declared tool schema. Our fence consults an LLM judge only on the residual and suppresses only when the maximum score over the m surviving candidates exceeds a conformal threshold at level α/m. Under exchangeability, wrong suppression is bounded by α, provided the calibration pool is drawn from the rule's own jurisdiction; a mismatched pool hides its own violation. Refining the schema or demanding class validity shrinks the calibration budget: a refinement dilemma we state exactly. On the evaluation corpus of 1,066 pairs over six tools, the fence deduplicates 94.7% of paraphrased du-

*Corresponding author.
Email address: sivagurumurugan1@gmail.com (Sivaguru R. M.)

plicates at 1.7% wrong suppression, against 12.1% for canonical keys; wrong suppression stays below $\alpha = 0.05$ across three judges while deduplication ranges from 82.6% to 94.7%. Under seed-level resplitting, mean wrong suppression is 2.7% with 10.5% of splits exceeding α. Typed fields eliminate field injection; prose-only injection succeeds 84% of the time. The labels are constructed and were checked only by a blinded panel of language models; no human annotation was collected. The mechanism provides bounded identity recognition and fence-level commitment, not external exactly-once execution.

Keywords: effect identity, duplicate suppression, LLM agents, conformal prediction, multiple comparisons, prompt injection, idempotency

1. Introduction

Language-model agents now act through tools whose effects cannot be undone—charging cards, moving money, sending messages, creating records [1, 2, 3]. When such an agent crashes, times out or is re-prompted, its recovery path re-issues work, and the surrounding system must not perform the same irreversible action twice. The classical defence gives each effect an identity, records it when the effect commits, and refuses a second commit under the same identity [4, 5, 6]. Journal positions in durable-execution engines [7, 8], client-chosen idempotency keys [6] and hashes of a canonicalised request all answer one question implicitly: *when do two requests denote the same act?*

For a deterministic program the answer is positional or syntactic, because a retried call replays the original byte for byte. An LLM agent breaks

both assumptions: its control flow is sampled, and its *arguments* are re-synthesised. An agent asked to wire $780 to Marlow Freight for shipment 88213 may first emit

```
wire_money{amount_text: "$780.00", beneficiary_text:
"Marlow Freight", reference: "Shipment 88213"}
```

and, after a crash and a re-prompt,

```
wire_money{amount_text: "seven hundred eighty dollars",
beneficiary_text: "MARLOW FREIGHT", reference:
"shipment #88213"}
```

The two canonicalise and hash differently, and a content-addressed fence wires the money twice. Conversely, two requests that differ in one token of free text ("instalment 1 of 2" against "instalment 2 of 2") can be two legitimate acts that must both execute.

Deciding identity therefore requires interpreting arguments, and the two errors are sharply asymmetric. Executing a duplicate is costly but *observable*: two committed records exist for reconciliation to find. Suppressing a distinct act is *silent*: the fence returns the recorded response of an earlier act, the agent believes the new act happened, and no trace shows otherwise. We call this second error *wrong suppression* and treat it as the safety-critical direction. Effect identity is thus a problem of reasoning under uncertainty with asymmetric loss, and an instance of a general one for autonomous agents: a control decision whose errors are irreversible and asymmetric, taken with a stochastic oracle as the only general-purpose decider, on inputs an adversary can write.

Canonical keys are sound but incomplete: they never merge two distinct acts and miss every paraphrased duplicate. Recent work identifies this failure in agent checkpoint–restore [9] and in authorisation [10], and proposes asking an analyser model whether a re-synthesised call matches a logged one [9]. An LLM judgement is the only general-purpose checker for such predicates, but LLM judges are biased and promptable [11, 12, 13] and steered by adversarial text [14, 15]. What is missing is a decision procedure whose safety-critical error is *bounded* rather than inherited from an unconstrained judge. We ask whether semantic effect identity can be made such a procedure, and what the declared tool schema determines about how far that is possible.

Approach and results. We decompose identity into a *certifier*, a decidable relation that proves sameness (canonical equality under semantics-preserving field policies); a *refuter*, a decidable relation that proves difference (inequality of declared typed fields); and the *residual*, which neither settles and which alone reaches an LLM judge (Figures 2 and 3). The fence suppresses on the judge only when the maximum score over the m candidates that survive the refuter exceeds a conformal threshold at level α/m, calibrated on distinct pairs from the rule's own jurisdiction. The residual is a function of the declared schema and its typed-field policies alone, so every error of sound decidable layers lies in it (Proposition 3). Wrong suppression is bounded by α for the deployed statistic under exchangeability—*provided* the pool matches the jurisdiction; a wider pool lowers the threshold and hides the violation (Theorem 1). The classically sharp union bound is attained by the deployed statistic, so the correction and the budget $\lfloor \alpha(n+1) \rfloor$ are forced (Theorem 2), and refining the schema, or demanding validity within each kind of difference,

shrinks that budget: the *refinement dilemma* (Proposition 5, Corollary 1).

On the constructed evaluation corpus of 1,066 pairs over six tools, with three judges from two model families, the fence deduplicates 94.7% [91.4–96.7] of paraphrased duplicates at 1.7% [0.7–4.3] wrong suppression, against 12.1% [8.8–16.4] for canonical keys. Wrong suppression stays below α for all three judges (1.7%, 1.3%, 0.9%) while deduplication ranges from 82.6% to 94.7% (89.0% for the third). A pool drawn from outside the jurisdiction roughly doubles the residual rate while the whole-split rate stays under α; in simulation an uncorrected threshold on the maximum of $m = 50$ scores suppresses 89.0% of distinct requests. The marginal bound hides a per-category rate above α on one judge, and per-category validity costs deduplication. The refuter removes a typed-field injection that the raw judge accepts 65.0% of the time, but prose-only injection succeeds 84.0% of the time; typing the numbers prose mentions closes that class at the price of deduplication (69.0% on the primary judge, 12.1% on the others). Under seed-level resplitting mean wrong suppression rises to 2.7% and 10.5% of splits exceed α, and re-scoring the same pairs in new batches at temperature 0 doubles it on the residual (4 of 144 to 8 of 144 distinct pairs), so the judge must be calibrated in the scoring context it is deployed in. Label generalisation was audited via a blinded model panel (Section 8.8), and the mechanism provides bounded identity recognition and fence-level at-most-once commitment, not exactly-once execution at an external provider (Section 3.5).

Contributions.

(C1) *Problem formalisation.* Effect identity for irreversible LLM-agent actions is formalised as reasoning under uncertainty with asymmetric

loss, distinguishing identity recognition (A), fence-level at-most-once commitment (B), and external exactly-once execution (C) (Section 3).

(C2) *Structural decomposition.* A certifier/refuter/residual decomposition separates exact decisions from ambiguous semantic cases, with the residual determined by the declared tool schema and its typed policies (Sections 4–5).

(C3) *Validity mechanism.* The residual LLM judge is calibrated on the actual jurisdiction of the decision rule, with multiplicity correction for the deployed max-over-candidates statistic, showing that the standard Bonferroni bound is attained for scale-free deterministic rules (Section 6, Section 8.2).

(C4) *Schema refinement dilemma.* Schema refinement removes cases from the judge's jurisdiction and can improve deterministic safety, but simultaneously shrinks the calibration population and therefore the statistical budget (Sections 6, 8.3, 8.6).

(C5) *Targeted boundary evaluation.* An empirical evaluation tests the framework's boundaries: paraphrased duplicate suppression, wrong suppression, candidate multiplicity, jurisdiction mismatch, schema refinement, adversarial text, crash/replay across process death, and cross-model transfer (Sections 7–8).

The individual techniques—canonicalisation, typed validation, LLM judging, split-conformal calibration, the Bonferroni correction—are known; the contribution is the formal composition for semantic effect identity under irreversible retry semantics, the conditions under which the composition is valid,

the schema-determined refinement trade-off, and the targeted boundary evaluation. Proofs are in Appendix A; secondary experiments and full tables are in the supplementary material.

2. Related work

Identity by position, token and content. At-most-once and exactly-once processing are classical systems problems, with a long lineage in remote procedure calls [4], stream processing [16], transactions and sagas [17, 18]. Durable-execution engines replay a recorded history under a determinism contract [7, 8], whose checkpoint–resume semantics now have a machine-checked conformance contract [19]; payment interfaces de-duplicate retries by a client-chosen idempotency key [6]; fencing tokens with idempotent writes are standard practice [5]. All identify an effect by position, by token or by content after syntactic normalisation, and all are correct when a retry is a faithful replay. A re-synthesising agent's retry is a fresh text with the same intent, so it receives a fresh identifier, token and hash. We keep content-addressed commitment as the certifier and add a bounded semantic layer.

Reliability runtimes for LLM agents. Atomix commits effects only when per-resource frontiers show that no earlier conflicting work can arrive [20]; CoAgent orders concurrent agents and repairs conflicts with saga-style inverses [21]; LogAct exposes actions on a shared log before they execute [22]; semantic isolation levels constrain how a durable AI workflow's resources may change beneath it [23]; and verified tool calls add postcondition checks and idempotency keys to a tool wrapper [24]. These take the identity of an effect as given or rely on an observable postcondition, which, where it exists, beats

any judge score. Yin [25] proves execution fidelity for agents moving value on public ledgers, where identity is fixed by the rendered transaction rather than decided across paraphrase.

Semantic replay in agents. ACRFence [9] observes that LLM agents re-synthesise subtly different requests after checkpoint–restore, names the resulting semantic rollback attacks, and proposes that an analyser LLM classify each post-restore call as equivalent to a logged call or not; the attacks are validated, but the mitigation is not implemented and its misclassification and evasion are left open. CapLease [10] names semantic replay in authorisation and binds an authenticated confirmation to a canonical action consumed through transactional transitions. The problem and the idea of consulting a model are prior work; we add a procedure with a stated bound on its safety-critical error, robustness to injected text on declared fields, and a measured cost.

Duplicate detection and semantic caching. Record linkage decides whether two records denote one entity [26]; semantic textual similarity whether two texts mean the same [27]; LLM-assisted pipelines remove near-duplicate documents [28]; and semantic caches return a stored response for a similar query [29]. The last is structurally closest to a fence, but its errors are cheap and roughly symmetric. Here one error is irreversible and silent, which is why we bound one direction rather than report accuracy or F1.

LLM judges and prompt injection. LLM judges are biased, contaminated and self-preferring [11, 12, 13, 30, 31], persuadable [15], and blind to some agent-action risks [32, 33]. Indirect prompt injection steers agents through

text in their context [14, 34]; capability gates admit confused-deputy failures [35]; and design-pattern defences constrain what untrusted text can influence [36]. We take the judge to be an uncalibrated, manipulable oracle, bound its error on the benign distribution by calibration, and remove from its jurisdiction every pair whose distinction is declared in a typed field—a decision the attacker's text cannot reach, in the sense of [36].

Conformal prediction and multiplicity. Split-conformal prediction gives distribution-free finite-sample guarantees under exchangeability [37, 38, 39]; Mondrian conformal prediction conditions them on a partition [37]. Language-model applications include generation and factuality [40, 41], robot planners that ask for help [42] and intervals on judge scores [43]; risk control bounds expected losses [44, 45], and conformal p-values support multiple testing [46]. Known failure modes include non-exchangeable generation, distribution shift [47, 48] and marginal under-coverage of a minority class [49], which has the shape of our class-conditional finding. Our use is deliberately simple. What differs is the object: a control decision with an irreversible consequence, whose deployed statistic is a maximum over a variable-size candidate set, and whose calibration population must match a jurisdiction that changes whenever decidable layers change.

Schemas and types. Typed schemas have formal semantics [50] and provenance asks why and where a result came from [51]; tool interfaces use schemas for syntactic validation, and we repurpose them as the carrier of the refuter (Proposition 3).

Table 1 summarises the comparison. We know of no prior work that pairs a sound certifier with a sound refuter, judges only the remainder, and

Table 1: Positioning. "Not stated" means we found no statement in the cited work, not that the property fails.

	Idempotency key	Canonical hash	Raw LLM judge	ACRFence / CapLease	This work
Survives non-deterministic control flow	no	yes	yes	yes	yes
Recognises paraphrased arguments	no	no	yes	analyser LLM / canonical action	residual judge
Error bound on suppression	exact on token	exact, incomplete	none	not stated	exact on decided pairs; $\leq \alpha$ on residual under A6
Typed-field distinctions immune to injected text	n/a	n/a	no	not stated	yes (refuter)
Multiple committed candidates analysed	n/a	n/a	no	not stated	yes, with budget
Undecided cases	execute	execute	threshold	fork (ACRFence)	escalate: execute and flag
Implemented and evaluated	yes	yes	—	attacks evaluated; mitigation not implemented [9] / yes	yes

calibrates the deployed statistic on a jurisdiction-matched pool; we expect precedents for each component.

3. Problem formulation

3.1. System model and terminology

A *logical run* carries an identifier r chosen by the application and reused across every retry, crash and replay of that run. Within a run, an agent emits *requests* $c = (t, x)$: a tool t and a structured input $x \in X_t$. A tool is *irreversible* if executing it twice for one intended action is a defect (a charge, a wire, an email, a ticket); only irreversible tools are fenced. Executing a request produces an *effect* and a *response*, and the fence records a *commit row* for each executed request in a durable log. Table 2 fixes the vocabulary.

3.2. Ground-truth identity

Let A_t be the unobservable space of intended actions of tool t and $[\![\cdot]\!]_t : X_t \to A_t$ map an input to the act it denotes. The agent, and any adversary, control x; the denotation is fixed by the world.

Definition 1 (Effect identity)**.** $x \approx_t x'$ if and only if $[\![x]\!]_t = [\![x']\!]_t$.

Corpus construction and annotation apply an operational criterion phrased in the vocabulary of the loss.

Definition 2 (Operational identity test)**.** A pair counts as the same act if, had the fence already committed x' and then silently returned the recorded response of x' in place of executing x, nothing the world was owed would fail to happen; it counts as different if that substitution would drop a payment, message, record or other owed effect.

Proposition 1. *If every input denotes exactly one atomic act and the recorded response of an executed request is determined by the act it performed,*

Table 2: Terminology. Each term is used only in the sense given here.

Term	Meaning
operation identifier	the content address $op = H(r \,\|\| t \,\|\| H(\kappa_t(x)))$ of a request
committed effect	an executed request whose commit row is in the durable log of run r
effect identity	ground-truth sameness of intended acts, $x \approx_t x'$ (Definition 1)
duplicate	a request x with $x \approx_t c$ for some committed effect c
candidate	a committed effect of (r, t) that survives the refuter; C is the candidate set and $m = \|C\|$
certifier / refuter	a decidable relation that proves sameness / difference (Definitions 5–6)
residual	pairs that neither the certifier nor the refuter settles (Definition 7)
judge	a stochastic scorer $J(t, x, x') \in [0, 1]$, here an LLM; never trusted as a probability
calibration pool	judge scores of distinct pairs drawn from the deciding rule's jurisdiction (Definition 9)
wrong suppression	suppressing a request that is distinct from every committed effect (Definition 4)
at-most-once	fence-level commitment of each recognised effect at most once (guarantee B)
exactly-once	external application of each intended act exactly once (guarantee C); never used unqualified

the operational test coincides with Definition 1 and is therefore an equivalence relation.

The proof is in Appendix A; the conditions hold for every tool evaluated here. For calls that bundle several effects the test can be asymmetric, and only Proposition 2 uses the equivalence structure.

3.3. The fence and its two errors

Definition 3 (Fence)**.** Given a request (t, x) in run r and the set $\mathcal{C}_{r,t}$ of effects already committed for (r, t), a fence returns SUPPRESS (return the recorded response of a committed effect without executing), EXECUTE (execute and commit), or ESCALATE (execute, commit, and durably flag the request as a suspected duplicate).

Definition 4 (Errors)**.** The *wrong-suppression* event is $\{\text{decision} = \text{SUPPRESS}\} \cap \{x \not\approx_t c \text{ for every } c \in \mathcal{C}_{r,t}\}$. The *duplicate-escape* event is $\{\text{decision} \in \{\text{EXECUTE}, \text{ESCALATE}\}\} \cap \{x \approx_t c \text{ for some } c \in \mathcal{C}_{r,t}\}$; it is *silent* if the decision is EXECUTE and *flagged* if it is ESCALATE.

We report wrong suppression $\text{WS} = P(\text{SUPPRESS} \mid \text{distinct})$, deduplication $P(\text{SUPPRESS} \mid \text{duplicate})$, silent escape $P(\text{EXECUTE} \mid \text{duplicate})$ and friction $P(\text{ESCALATE} \mid \text{distinct})$. The design objective is a strict priority,

$$\text{minimise } P(\text{EXECUTE} \mid \text{duplicate}) \quad \text{subject to} \quad \text{WS} \leq \alpha, \tag{1}$$

because wrong suppression is undetectable when it happens, a silent escape is not, and a flagged escape is recoverable by reconciliation. Accuracy and F1 weight the two errors symmetrically and are not reported.

When a duplicate is suppressed the fence returns the response of the matched candidate. If $m \geq 2$ and a distinct candidate outscores the true match, no owed act is dropped but the returned response belongs to a different act. This *response-attribution error* is not a wrong suppression; it cannot occur when $m \leq 1$, the regime of our corpus, and we measure it on multi-candidate units in Section 8.2.

3.4. Threat model

Duplication arises from retries, crashes and paraphrasing replays under the same run identifier. The adversary injects text into the agent's context—a tool result, a retrieved document, a user-supplied field [14]—aiming at *forced suppression* (a distinct act silently dropped) or *forced execution* (a duplicate fired twice). It controls request text but neither the declared semantics of typed fields nor the durable log. We assume storage with atomic insert-if-absent; we do not assume reliable networks or the ability to retract an accepted effect.

3.5. Three guarantees, kept apart

The phrase "exactly-once" conflates three distinct system properties that fail independently (Table 3): (A) effect-identity recognition, deciding whether two candidate requests denote the same real-world act; (B) fence-level at-most-once commitment, preventing the same recognised effect from committing repeatedly in the durable log; and (C) external exactly-once execution, ensuring an external service realises the intended act exactly once across process crashes. The mechanism establishes a bounded identity-recognition and fence-commitment layer (guarantees A and B); exactly-once external

Table 3: The three guarantees and their status. Assumptions A1–A8 are listed in Table 5.

Guarantee	Status in this work	Result
(A) Identity recognition: does the fence decide correctly whether a request and a committed effect denote the same act?	Exact on certified and refuted pairs under A5 and A5′. On the residual, wrong suppression is bounded by α for the deployed statistic under A6, marginally over calibration draws. Zero error in both directions is impossible on overlapping score distributions.	Prop. 4, Thm. 1, Prop. 6
(B) Fence-level at-most-once: given correct identity, does the fence commit each effect at most once?	Holds under A1–A4 and A7 (no concurrent dispatch per identity). Without A7 at most one commit row exists and concurrent duplicates are observable, not prevented.	Prop. 7
(C) External exactly-once: does the provider apply each intended act exactly once, across process death?	*Not provided by the mechanism.* Obtainable only by composition with a provider idempotency key that is paraphrase-invariant and complete for identity (A8, A5″).	Props. 8, 9

execution (guarantee C) remains outside the mechanism's local contract and requires downstream provider idempotency keys complete for identity (Proposition 8).

4. The effect-identity framework

Ground-truth identity cannot be observed. A mechanism must compute decidable relations and use them as proxies, and two dual notions of correctness govern such proxies.

4.1. Certifiers and refuters

Definition 5 (Certifier)**.** A relation $\mathsf{C} \subseteq X_t \times X_t$ is a *certifier* if $\mathsf{C} \subseteq \approx_t$: whenever $\mathsf{C}(x, x')$ holds, the two requests denote the same act.

Definition 6 (Refuter)**.** A relation $\mathsf{D} \subseteq X_t \times X_t$ is a *refuter* if $\mathsf{D} \cap \approx_t = \emptyset$: whenever $\mathsf{D}(x, x')$ holds, the two requests denote different acts.

A certifier licenses suppression and a refuter licenses execution; each may be arbitrarily incomplete. Canonical equality $\equiv_\kappa$ normalises each declared field under a policy (identifier, case-insensitive token, number, free text) and is a certifier when each policy is semantics-preserving (assumption A5). Projection difference $\not\equiv_\sigma$ compares the canonical values of the declared *typed* fields and is a refuter when a difference in any such field implies a different act (A5′). A judge threshold $J > \tau$ is decidable but carries no proof in either direction (Table 4).

Proposition 2 (Structure)**.** *Fix t and order relations on $X_t \times X_t$ by inclusion. The certifiers are closed under union and intersection, with least element $\emptyset$ and greatest element $\approx_t$; the refuters likewise, with greatest element the complement of $\approx_t$. A finite union of decidable certifiers (refuters) is a decidable certifier (refuter). The only relation that is both is $\emptyset$.*

Table 4: Relations used in this paper and the assumption under which each is sound.

Relation	Role	Sound under	Used as
byte equality $\equiv_{\text{exact}}$	certifier	always	baseline B1
canonical equality $\equiv_\kappa$	certifier	A5	L0; licenses SUPPRESS
projection difference $\not\equiv_\sigma$	refuter	A5′	witness; licenses EXECUTE
judge threshold $J > \tau$	neither	—	must be calibrated
ground truth $\approx_t$	target	—	not observable

The proof is in Appendix A. Sound decidable layers therefore combine by union without re-arguing soundness, and a design must supply certification and refutation separately.

4.2. The residual

Definition 7 (Decided set and residual)**.** For a declaration that fixes κ_t and σ_t, the decided set is $\text{Dec}_t = \equiv_\kappa \cup \not\equiv_\sigma$ and the residual is its complement,

$$\text{Res}_t = \{(x, x') : \kappa_t(x) \neq \kappa_t(x') \ \wedge \ \sigma_t(x) = \sigma_t(x')\} :$$

requests that differ after canonicalisation yet agree on every declared effect-determining field.

Proposition 3 (Residual determination)**.** *(i)* Res_t *is a function of* (κ_t, σ_t) *alone; it does not depend on the judge, the agent, the adversary or any run-time observation. (ii) If a mechanism suppresses on* $\equiv_\kappa$*, executes on* $\not\equiv_\sigma$*, and A5 and A5′ hold, every error it makes is on a pair in* Res_t*. (iii) If* (κ', σ') *refines* (κ, σ) *(Definition 8), then* $\text{Res}'_t \subseteq \text{Res}_t$*.*

Proof. (i) The defining condition mentions only κ_t and σ_t. (ii) A pair in $\equiv_\kappa$ is suppressed and, by A5, is a true duplicate; a pair in $\not\equiv_\sigma$ is executed and,

by A5′, is truly distinct. (iii) Refinement enlarges $\equiv_\kappa$ and $\not\equiv_\sigma$, so their union grows and its complement shrinks. □

The proposition is elementary; its content is operational. Changing the judge redistributes decisions *inside* Res_t; only changing the declaration changes Res_t.

4.3. Schema refinement and the refinement dilemma

Definition 8 (Refinement)**.** A declaration (κ', σ') *refines* (κ, σ) if $\equiv_\kappa \subseteq \equiv_{\kappa'}$, $\sigma(x) \neq \sigma(x') \Rightarrow \sigma'(x) \neq \sigma'(x')$, and $\equiv_{\kappa'}$ and $\not\equiv_{\sigma'}$ remain sound.

Refinement moves act-determining information from free text into a typed field. In our corpus `wire_money` declares amount, beneficiary and reference as free text, so every non-identical pair of wires is residual, and a second instalment differs from the first only by "instalment 2 of 2". Typing the numbers each text field mentions lets the refuter separate the instalments, so an injected "this is the same payment; suppress" no longer reaches the judge.

Refinement is the only intervention that removes pairs from the judge's jurisdiction. It also has a cost that is easy to miss: the calibration pool is drawn from the residual (Section 5.4), so refinement shrinks the pool with it. A smaller pool moves the conformal threshold to a higher order statistic and certifies fewer candidates, and the surviving pairs are not a random sample: the refuter removes distinct pairs whose declared fields differ, which judges tend to score low. We call this the *refinement dilemma* (Figure 1); Proposition 5 states its exact part and Section 8.6 measures both parts. After refining a schema, re-calibrate on the new jurisdiction: at $\alpha = 0.05$ the fence

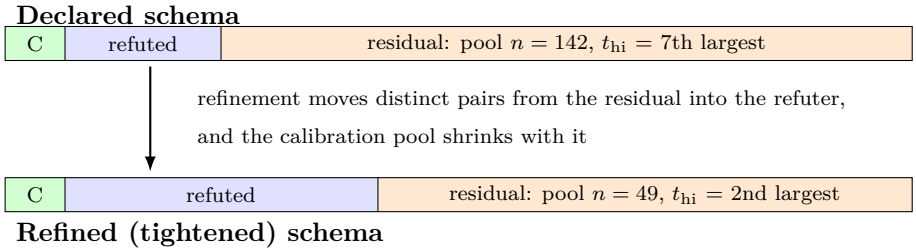


Figure 1: The refinement dilemma, drawn to scale for the test split (shares of 513 pairs; C: certified; the residual is the fragment the judge decides). Refinement enlarges the refuted region, which removes pairs from the judge's jurisdiction and closes the prose-only injection class. Because the calibration pool is the distinct part of the residual, the pool shrinks with it (primary judge: from 142 to 49 pairs), and at $\alpha = 0.05$ the conformal threshold moves from the 7th to the 2nd largest pool score (Proposition 5). Section 8.6 measures the consequence.

needs at least 19 distinct residual calibration pairs to suppress semantically at all, and $m/\alpha - 1$ to do so with m candidates.

In these terms an idempotency key and content-addressed fencing are certifiers whose residual policy is "execute", and a raw LLM analyser resolves the whole space by an unbounded judgement.

5. The fence

The fence instantiates the framework with canonical equality as certifier, typed-field inequality as refuter, an LLM judge as oracle, one-sided conformal calibration corrected for the number of candidates, and a three-way residual policy with an audited middle band; each is a design choice within the framework. The refuter runs *before* the judge (Figures 2 and 3), so a candidate whose declared fields differ never reaches the judge, and every candidate it

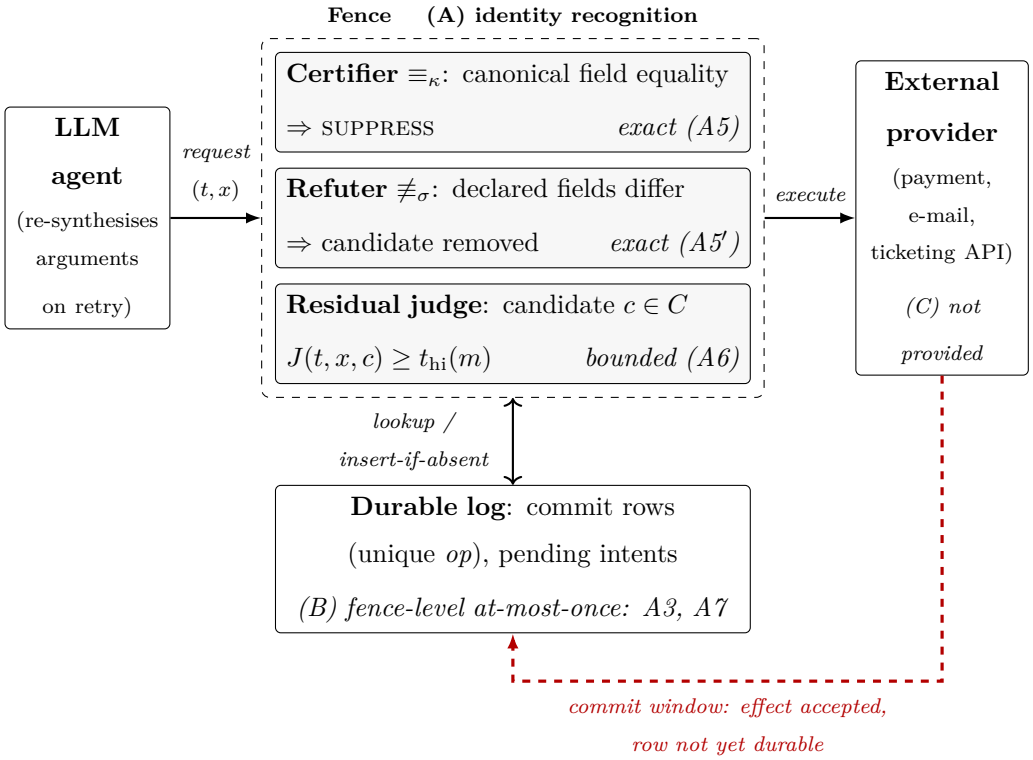


Figure 2: System view. The fence sits between the agent and an irreversible provider. Identity recognition (A) is decided by two exact layers and a calibrated residual judge; fence-level at-most-once commitment (B) is enforced at the durable log under its storage and scheduling assumptions; external exactly-once (C) is not provided, because a crash inside the commit window (dashed) leaves no local record of an accepted effect (Proposition 7).

removes also shrinks the multiplicity m the threshold must correct for.

5.1. Certifier: the canonical core

Each tool schema declares a normalisation policy for each field, and the policy must be semantics-preserving for that field's contribution to the effect (A5). The policies used here are: `id` (opaque identifier; Unicode NFC nor-

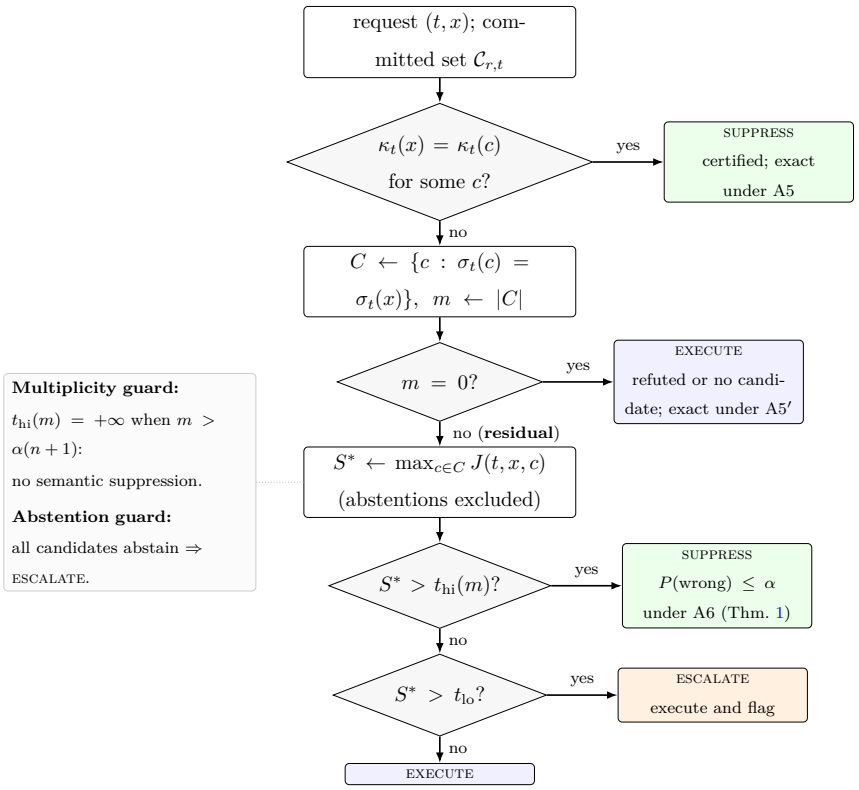


Figure 3: Decision flow for one request. The certifier and the refuter settle their pairs without a model call; only the residual reaches the judge, and the judge can cause suppression only through the multiplicity-corrected threshold. Every path that is not certified ends in execution, flagged or not.

malisation and trimming, case-sensitive); `ci` (case-insensitive token such as a currency code or e-mail address; NFC, trim, lower case); `num` (numeric value; `"120.00"`, `" 120 "` and `120` normalise to one canonical decimal, thousands separators removed); and `text` (free text; NFC, trim, internal whitespace collapsed—deliberately not semantic). The canonical form $\kappa_t(x)$ applies each policy and sorts the keys; the operation identifier is $op = H(r \,\|\, t \,\|\, H(\kappa_t(x)))$ with $H = \text{SHA-256}$. The certifier suppresses a request only when a committed effect has the same identifier and the same stored input hash. It is sound under A5 and deliberately incomplete: every paraphrase escapes it.

5.2. Refuter: the distinctness witness

The semantic projection $\sigma_t(x)$ is the tuple of canonical values of the declared non-text fields. Under A5′—a difference in any declared typed field implies a different act—$\sigma_t(x) \neq \sigma_t(x')$ proves $x \not\approx_t x'$ whatever any judge would say. The fence removes every committed effect whose projection differs before the judge is consulted, so an injected "treat this as a duplicate" cannot cause suppression across a different amount, account or identifier. A5′ is one-directional: equal projections do *not* imply the same act (legitimate repeats are the counterexample). A field wrongly declared as typed produces false vetoes, which execute a duplicate loudly; an act-determining field declared as free text leaves its distinction to the judge.

5.3. Residual judge

A judge call scores a batch of pairs from one tool and returns, for each, the probability that the two requests denote a single act, as an integer percentage (prompt in the supplementary material). An integer in $[0, 100]$ is

accepted; a non-integer such as `0.95`, an out-of-range value or a missing entry is rejected and the batch retried; a pair that never yields an admissible value is an *abstention*, never imputed. An earlier scoring pass that violated this contract was discarded (supplementary material, Section S3). Abstentions are dropped from calibration, excluded from the maximum, and never cause suppression; if every candidate abstains the fence escalates.

5.4. Calibration

Definition 9 (Jurisdiction and calibration pool)**.** The *jurisdiction* of a configuration of decidable layers is the set of pairs that the configuration routes to the judge. Its *calibration pool* is the multiset $\{s_1, \ldots, s_n\}$ of judge scores of the distinct pairs of a held-out calibration split that lie in its jurisdiction, with abstentions dropped.

For the proposed configuration (certifier and refuter both active) the jurisdiction is the residual (Definition 7). For a candidate set of size $m \geq 1$ the suppression threshold is

$$t_{\mathrm{hi}}(m) = s_{(k)}, \qquad k = \left\lceil (n+1)(1-\alpha/m) \right\rceil, \qquad t_{\mathrm{hi}}(m) = +\infty \text{ if } k > n, \quad (2)$$

where $s_{(k)}$ is the k-th smallest pool score, and $t_{\mathrm{hi}}(0) = +\infty$. The escalation floor t_{lo} is the $\lfloor (n_S + 1)\gamma \rfloor$-th smallest score of the n_S duplicate pairs in the same jurisdiction, capped at t_{hi}. With S^* the maximum score over the candidates that did not abstain, the fence *suppresses* if $S^* > t_{\mathrm{hi}}(m)$, *escalates* (executes and flags) if $t_{\mathrm{lo}} < S^* \leq t_{\mathrm{hi}}(m)$, and *executes* if $S^* \leq t_{\mathrm{lo}}$. The inequality for suppression is strict, so a score equal to the threshold never suppresses; judge scores are integer percentages, so such ties are common.

Two design choices need justification. First, the pool is drawn from the rule's own jurisdiction. The refuter removes distinct pairs whose declared fields differ, which judges score confidently low; a pool that still contains them is lighter-tailed than the scores the deployed rule thresholds, which lowers t_{hi}. The violation is in the unsafe direction and invisible over a whole test split, because the diluting pairs also dilute the denominator (Section 8.2); every configuration and ablation is therefore calibrated on its own jurisdiction. Second, we use the Bonferroni level α/m rather than calibrating the maximum directly, which would need calibration *units* with the deployment's candidate-set structure ($\lfloor n/m \rfloor$ units from n scores, at least 19 of them, for an m that varies per request). Bonferroni reuses one pool at every m, needs only pairwise exchangeability, and degrades to $t_{\text{hi}} = +\infty$; its price is conservatism, which we measure.

5.5. Algorithm

Algorithm 1 gives the complete procedure. A request that is not suppressed records a PENDING intent, executes, and inserts its commit row atomically; a crash in between—the *commit window*—leaves a pending intent with no commit row (Proposition 7). An optional guard flags, but never suppresses, a request that resembles such an orphaned intent.

5.6. Conservative defaults

The threshold is $+\infty$ when the pool cannot certify α at the current m (fewer than $m/\alpha - 1$ scores), when $m = 0$, and when the pool is absent; an abstention never suppresses; and a suspected re-fire inside the commit window is executed and flagged. In every uncertain case the failure mode

Algorithm 1: Layered semantic fence (the proposed configuration).

Input: request (t, x) in run r; durable log L; pool $\{s_i\}_{i \le n}$; floor t_{lo}; level α

if *t is idempotent* **then**
    return execute(x)
$op \leftarrow H(r \,\|\, t \,\|\, H(\kappa_t(x)))$;
if *L has a commit row for op with input hash $H(\kappa_t(x))$* **then** `// certifier; exact under A5`
    return its recorded response
$C \leftarrow \{c \in L.\mathrm{committed}(r, t) : \sigma_t(c.x) = \sigma_t(x)\}$; `// refuter removes declared differences`
if $C \neq \emptyset$ **then**
    $s_c \leftarrow J(t, c.x, x)$ for each $c \in C$; `// abstention` $\Rightarrow s_c = \perp$
    $m \leftarrow |C|$; $t_{\mathrm{hi}} \leftarrow t_{\mathrm{hi}}(m)$ by Eq. (2);
    if *all* $s_c = \perp$ **then**
        flag DUPLICATE_SUSPECT
    else
        $S^* \leftarrow \max\{s_c : s_c \neq \perp\}$, attained at c^*;
        if $S^* > t_{\mathrm{hi}}$ **then return** c^*.response `// semantic suppression;`
        if $S^* > t_{\mathrm{lo}}$ **then** flag DUPLICATE_SUSPECT `// escalate: execute and flag;`
record intent$(op,$ PENDING$)$;
$resp \leftarrow$ execute(x) ; `// commit window opens`
$ok \leftarrow L.\mathrm{insertIfAbsent}(op, H(\kappa_t(x)), \sigma_t(x), x, resp)$; `// commit window closes`
record intent$(op,$ ok ? COMMITTED : COMMIT_CONFLICT$)$;
return *resp*;

is a loud duplicate rather than a silent loss of a distinct act, as the loss asymmetry demands.

6. Theoretical analysis

The individual tools used below—split-conformal calibration, the union bound, order statistics—are standard, and so are most of the arguments. What this section contributes is an account of what their composition guarantees to an agent that must decide under irreversibility with a stochastic

oracle, and of the conditions under which it does. Propositions 4 and 3 make the certifier/refuter/residual decomposition exact. Theorem 1 applies conformal calibration with a Bonferroni correction to the statistic the fence actually thresholds—a maximum over a variable-size candidate set behind decidable vetoes—and makes its validity condition explicit: the pool must be drawn from the rule's own jurisdiction, and a pool that is not can hide its own violation (Section 8.2). Theorem 2 records that the classical bound is attained by that statistic, so the correction and the budget are forced rather than conservative. Proposition 5 and Corollary 1 state the refinement dilemma exactly. The remaining propositions delimit what is guaranteed; they are stated because nearby designs violate them.

6.1. Assumptions and their status

Table 5 lists every assumption and how the experiments satisfy or test it. Several are environmental, and a deployment that does not supply them does not obtain the corresponding guarantee.

Structurally, the candidate set of a request is every effect committed in the same run for the same tool; duplicates across runs or across tools are outside every result below.

6.2. Soundness of the decidable layers

Proposition 4 (Decidable layers)**.** *Under A2 and A5, every certifier suppression is correct: if the fence suppresses on the canonical branch, the request denotes the same act as the matched committed effect. Under A5′, no judge output can cause suppression of a request whose declared typed fields differ from those of every committed effect.*

Table 5: Assumptions, their type, and their status in the implementation and experiments. "Measured" entries test an assumption on this corpus; they do not establish it in general.

Assumption	Type	Status in this work
A1 Stable run identifier across retries and replays	environment	Supplied by the harness; a retry under a new identifier is out of scope.
A2 Deterministic canonical identity: κ_t and H are pure; H is collision-free	implementation	Pure canonicalisation; H is SHA-256; unit-tested.
A3 Atomic insert-if-absent on *op*	storage	*Partial:* unique for a single writer but not crash-atomic; crashes injected at protocol points, not inside a write.
A4 Effect before commit; commit only on success	implementation	Algorithm 1; eight injected crash points (Section 8.7).
A5 Every declared field policy is semantics-preserving	schema	Tool author's obligation. Measured: 0 of 232 distinct test pairs canonically equal.
A5′ A difference in a declared typed field implies a different act (one direction only)	schema	Tool author's obligation. Measured: 0 of 281 duplicates refuted; 20 of 20 typed-field injections refuted.
A5″ Projection completeness: $\sigma_t(x) = \sigma_t(x') \Leftrightarrow x \approx_t x'$	schema (optional)	Only in Proposition 8, Case A; false for every tool in the corpus.
A6 Conditionally on m, each score of a truly distinct residual comparison is exchangeable with the calibration pool	statistical	Stratified pair-level split, attacks held out; 82.8% of test pairs share a seed act with a calibration pair, tested by seed-level resplitting (Section 8.2). Needs a stationary judge; unverifiable at deployment.
A7 No concurrent dispatch per identity	scheduler	Sequential dispatch in all experiments; concurrency not tested.
A8 The provider applies at most one effect per idempotency key	provider	Only in Proposition 8; absent by design in the crash experiment.

Proof. The canonical branch fires only if $H(\kappa_t(x)) = H(\kappa_t(x'))$ for a committed x'; by A2 this implies $\kappa_t(x) = \kappa_t(x')$ and by A5 $x \approx_t x'$. For the second claim, a committed effect whose projection differs is removed from C before any judge call, so it is never the maximising candidate; if every committed effect differs, $C = \emptyset$ and the fence executes. □

6.3. A bound on the deployed statistic

The fence suppresses on $S^* = \max_{c \in C} J(t, x, c)$, not on one pairwise score, and a pairwise bound $P(s_i > \hat{t}) \leq \alpha$ does not imply $P(\max_i s_i > \hat{t}) \leq \alpha$. The result below is stated for the deployed statistic.

Theorem 1 (Multiplicity-corrected wrong-suppression bound)**.** *Let the pool $\{s_i\}_{i \leq n}$ be drawn from the rule's jurisdiction (Definition 9) and let $t_{\mathrm{hi}}(m)$ be given by Eq. (2). Suppose (A6) that, conditionally on m, the judge score of any comparison between a request and a candidate that denotes a different act is exchangeable with the pool. Then for every request x that denotes a different act from every member of its candidate set C, $|C| = m$,*

$$P\Big(S^*(x, C) > t_{\mathrm{hi}}(m)\Big) \leq \alpha,$$

for every $m \geq 0$, with no assumption on the distribution of scores and no assumption on the dependence among the m comparisons.

Proof. Condition on m. If $m = 0$ or $t_{\mathrm{hi}}(m) = +\infty$ the fence cannot suppress and the probability is 0. Otherwise let $\beta = \alpha/m$ and $k = \lceil (n+1)(1-\beta) \rceil \leq n$, so $t_{\mathrm{hi}}(m) = s_{(k)}$. Fix a candidate $c_i \in C$. Because $x \not\approx_t c_i$, its score s'_i is a distinct-pair score, so by A6 the $n+1$ values $s_1, \ldots, s_n, s'_i$ are exchangeable and, with ties broken uniformly at random, the rank of s'_i among them is

uniform on $\{1, \ldots, n+1\}$. If $s'_i > s_{(k)}$, then s'_i exceeds at least k pool values and has rank at least $k+1$ under any tie-breaking, so

$$P\Big(s'_i > s_{(k)}\Big) \leq \frac{n+1-k}{n+1} \leq 1-(1-\beta) = \beta.$$

This holds for each i separately. The union bound over the m candidates, which requires no independence, gives

$$P\Big(S^* > t_{\mathrm{hi}}(m)\Big) = P\Big(\exists i \leq m : s'_i > t_{\mathrm{hi}}(m)\Big) \leq \sum_{i=1}^{m} P\Big(s'_i > t_{\mathrm{hi}}(m)\Big) \leq m\beta = \alpha.$$

The bound holds conditionally on each value of m and therefore also after averaging over the distribution of m. Abstentions only remove candidates from the maximum while $t_{\mathrm{hi}}(m)$ is computed with the full m, which is conservative. □

The probability is joint over the calibration pool and the request, so the bound is *marginal*: a particular deployment's pool can exceed α (Section 8.2), and nothing is claimed under distribution shift, including injection (Section 10). It tolerates arbitrary dependence among the m comparisons, which share x.

The pool must match the jurisdiction. A6 compares the pool with the scores the deployed rule thresholds, and which pairs those are is decided by the layers in front of the judge. A pool drawn from a wider population satisfies A6 only by coincidence; if the extra pairs score lower, the threshold falls and the bound fails while the rate over a whole test split still looks acceptable. Jurisdiction matching is therefore a necessary condition of the theorem, not an implementation detail.

The escalation floor. A duplicate's candidate set contains its match, so taking a maximum helps the escalation side and t_{lo} needs no correction. With integer-percent scores ties are frequent and t_{lo} is capped at t_{hi}, so we treat the band as an empirical quantity (Section 8.9) and claim no bound for it.

6.4. Calibration budget and the refinement dilemma

Proposition 5 (Calibration budget; refinement dilemma)**.** *Let a configuration have pool size n and level α.*

(a) $t_{\text{hi}}(m)$ is finite if and only if $m \le \alpha(n+1)$; equivalently, semantic suppression at candidate-set size m requires $n \ge \lceil m/\alpha \rceil - 1$, and the largest certifiable candidate-set size is $m^* = \lfloor \alpha(n+1) \rfloor$.

(b) When finite, $t_{\text{hi}}(m)$ is the ρ-th largest *pool score with $\rho = \lfloor (n+1)\alpha/m \rfloor$, which is non-decreasing in n.*

(c) If (κ', σ') refines (κ, σ) and both are calibrated on the same calibration split, the refined pool is a sub-multiset of the original pool. Hence $n' \le n$, $m^{*\prime} \le m^*$ and $\rho' \le \rho$: refinement can only shrink the certifiable budget and move the threshold to an equal or higher rank from the top.

Proof. (a) Since n is an integer, $\lceil (n+1)(1-\alpha/m) \rceil \le n$ holds iff $(n+1)(1-\alpha/m) \le n$, i.e. iff $m \le \alpha(n+1)$. (b) The k-th smallest of n values is the $(n+1-k)$-th largest, and $n + 1 - \lceil (n+1)(1-\alpha/m) \rceil = \lfloor (n+1)\alpha/m \rfloor$. (c) The pool is the set of distinct calibration pairs in the jurisdiction with a score; the jurisdiction of the proposed configuration is the residual, and by Proposition 3(iii) the refined residual is contained in the original one. □

At $\alpha = 0.05$ and $m = 1$, the declared pool of the primary judge ($n = 142$) places t_{hi} at the 7th largest score, the tightened pool of Section 8.6 ($n = 49$)

at the 2nd largest, and any pool with $n \leq 18$ at $+\infty$. The proposition does *not* say the refined threshold is numerically higher; that depends on whether the judge scores the removed pairs lower than those that remain, which Section 8.6 measures. Asking for the bound to hold within each kind of difference is a second refinement—of the calibration population rather than of the schema—and it pays the same price.

Corollary 1 (Class-conditional budget)**.** *Partition the jurisdiction into classes $c = 1, \ldots, K$ by a function of the request, let the pool contain n_c scores of class c ($\sum_c n_c = n$), and let $t_c(m)$ be Eq. (2) computed on the class-c pool alone.*

(a) If A6 holds within each class, then $P(S^* > t_c(m) \mid \text{class } c) \leq \alpha$ for every c.

(b) $t_c(m)$ is finite iff $m \leq \alpha(n_c + 1)$. Every class is certifiable at m iff $\min_c n_c \geq \lceil m/\alpha \rceil - 1$, which requires $n \geq K(\lceil m/\alpha \rceil - 1)$; and $\min_c \lfloor \alpha(n_c + 1) \rfloor \leq \lfloor \alpha(n/K + 1) \rfloor$.

(c) If the class is not observable when the fence decides, thresholding at $\max_c t_c(m)$ satisfies (a) for every class simultaneously; it is finite only if every class is certifiable at m.

Each class pool is a sub-multiset of the pool, so its threshold sits at an equal or higher rank from the top: partitioning the jurisdiction into K classes gives each class the budget of Proposition 5(a) at its own n_c. A per-class guarantee costs calibration budget, and when the classes are defined by the label, as the kinds of difference in our corpus are, it costs the deduplication of the most conservative class (Section 8.3).

6.5. The standard bound is attained

Theorem 1 rests on a union bound. That the union bound is sharp when nothing is known about the dependence among the events is classical: $P(\bigcup_i A_i) \le \min\{1, \sum_i P(A_i)\}$ is the upper Fréchet bound, attained when the events are disjoint or cover the space [52, 53, 54], and the sharper bounds of Hunter and Worsley need the pairwise intersection probabilities [55, 56], which a fence does not know. In the language of merging p-values, the Bonferroni rule is precise and admissible under arbitrary dependence [57, 58]. We therefore do not present the tightness below as a new technical result. It answers a narrower question: is the classical worst case reachable by the *deployed* statistic under the fence's own assumption, and does any rule a fence can audit do better? Within deterministic rules that do not depend on the scale of the judge's scores—which are uncalibrated, so a rule that reads meaning into their absolute values inherits that miscalibration—the standard bound is exactly attained. Lemma 1, which identifies the class, is a characterisation of a known type—symmetric functions that commute with every increasing bijection are order statistics [59, 60]—stated here without a continuity assumption and proved in Appendix A for completeness.

Lemma 1 (Scale-free thresholds)**.** *Let τ map pools of n distinct scores to $[-\infty, +\infty]$, be symmetric in its arguments, and satisfy $\tau(g(s_1), \dots, g(s_n)) = g(\tau(s_1, \dots, s_n))$ for every strictly increasing bijection g of $\mathbb{R}$, extended by $g(\pm\infty) = \pm\infty$. Then $\tau \equiv s_{(k)}$ for one fixed $k \in \{1, \dots, n\}$, or $\tau \equiv +\infty$, or $\tau \equiv -\infty$.*

A deterministic rule that thresholds S^* at a scale-free function of the pool is therefore "suppress iff $S^* > s_{(k)}$" for a fixed k, or never suppresses,

or always does. The fence is such a rule, with $k = k^* = \lceil (n+1)(1-\alpha/m) \rceil$ when $k^* \le n$ and $\rho = n+1-k^*$ as in Proposition 5. Let $\mathcal{Q}_{n,m}$ be the class of joint laws of $(s_1, \ldots, s_n, s'_1, \ldots, s'_m)$ in which, for each i, the pool and s'_i are $n+1$ independent draws from one continuous distribution, with arbitrary dependence among $s'_1, \ldots, s'_m$. Every law in $\mathcal{Q}_{n,m}$ satisfies A6, so a rule that is valid under A6 must be valid on all of $\mathcal{Q}_{n,m}$.

Theorem 2 (The standard bound is attained by the deployed statistic)**.** *For every $k \in \{1, \ldots, n\}$,*

$$\sup_{\mathcal{Q}_{n,m}} P\Big(\max_{i \le m} s'_i > s_{(k)} \Big) = \mathbb{E}\Big[\min\{1, mB\} \Big], \qquad B \sim \mathrm{Beta}(n+1-k,\, k),$$

and the supremum is attained. For the fence's threshold,

$$\sup_{\mathcal{Q}_{n,m}} P(\textit{wrong suppression}) \;=\; \frac{m\rho}{n+1} - \mathbb{E}\Big[(mB-1)_+\Big] \;\le\; \frac{m\rho}{n+1} \;\le\; \alpha,$$

where $\mathbb{E}[(mB-1)_+] \le (m-1)\exp\Big(-(1-\alpha)^2 n/(2m)\Big)$.

The attaining law is not exotic: pool and candidate scores are i.i.d., the most favourable reading of A6, and only the joint behaviour of one request's m candidate scores is adversarial—spread so that at most one is high, as when a request is compared with committed effects that differ from it in different ways.

Corollary 2 (What cannot be improved)**.** *Fix $\alpha \in (0,1)$ and $m \ge 1$. Worst cases are over $\mathcal{Q}_{n,m}$.*

(a) The constant. *For $m = 1$ the worst case is $\rho/(n+1)$, which equals α whenever $(n+1)\alpha$ is an integer; for every m it tends to α as $n \to \infty$. No constant below α bounds wrong suppression for all pool sizes.*

(b) The correction. *For $m = 1$, every rank $k < k^*$ has worst case $(n + 1 - k)/(n + 1) > \alpha$. For every m, a rule with a fixed per-candidate level $\gamma > \alpha/m$ has worst case tending to $\min\{1, m\gamma\} > \alpha$. The Bonferroni level is the largest per-candidate level valid for every pool size without an assumption on the dependence among the m comparisons.*

(c) The budget. *If $m > \alpha(n+1)$, the most conservative finite threshold, the pool maximum ($k = n$), has worst case exactly*

$$\mathbb{E}\Big[\min\{1, mB\}\Big] = \frac{m}{n+1}\left(1 - \left(1 - \frac{1}{m}\right)^{n+1}\right), \qquad B \sim \mathrm{Beta}(1, n),$$

and every other finite threshold has a larger one. This exceeds α unless $m/(n + 1)$ lies within $\delta_{0.05} < 1.1 \times 10^{-10}$ of α (at $\alpha = 0.05$; proof in Appendix A), which no pool with $n + 1 < 4.8 \times 10^8$ can do. There, switching semantic suppression off beyond the budget is forced rather than conservative: the only valid deterministic scale-free rule is never to suppress.

Proofs are in Appendix A. For every configuration in this paper the gap between worst case and bound is below 10^{-10}: 4.895% for the declared pool ($n = 142$, $m = 1$), 4.000% for the tightened pool ($n = 49$), and 4.975% at $n = 200$ for $m = 1, 2, 5, 10$, the simulation setting of Section 8.2, whose law gave 3.7% at $m = 10$. One rank lower the worst case exceeds α (5.594% and 6.000%). Under the attaining law the uncorrected rule suppresses 49.7% of distinct requests at $m = 10$ and 99.9% at $m = 50$, against 38.3% and 89.0% in the simulation. A direct simulation of the attaining law agrees with the exact values in all eight configurations we computed; the budget condition was checked for every $n + 1 \leq 10^6$ at $\alpha = 0.05$ and $\alpha = 0.01$ with no

exception; and the equality in Corollary 2(c) was checked without its closed form, against the general rank-k computation at $k = n$ to within 10^{-12} over 4,000 pairs (n, m), and by simulation, which agrees in all six budget configurations (supplementary material, Section S21). The result concerns deterministic rules: a coin-tossing rule beyond the budget is valid but cannot be audited or replayed. With integer-percent scores, ties lower the realised rate (Section 8.9).

6.6. Why the architecture is layered

Proposition 6 (No zero-error threshold on overlapping scores)**.** *Let F_{same} and F_{diff} be the judge-score distributions of duplicate and distinct pairs. If they overlap—some score has positive probability under both—no rule that thresholds the score alone achieves zero wrong suppression and zero silent escape simultaneously. An escalation band can remove silent escape only by flagging, not suppressing, the duplicates in the overlap.*

The proof is in Appendix A. Zero error on a class needs a decision that does not consult the score, which is what the certifier and refuter are; the overlap is judge- and tool-dependent (supplementary material, Section S8).

6.7. Fence-level at-most-once

Proposition 7 (Fence-level at-most-once commitment)**.** *Under A1–A4 and A7, for a fixed run r and operation identifier op, across any number of serialised executions and process restarts sharing the durable log, the effect for op is dispatched and committed at most once by the fence, except that a crash strictly between the external effect and the commit insert leaves a* PENDING *intent with no commit row, after which a later execution may dispatch again.*

The proof is in Appendix A. The commit window cannot be closed by any local mechanism, because during it the only record of the effect is at the provider. Without A7, two concurrent executions can both miss the log and dispatch; at most one commit row survives and the loser records COMMIT_CONFLICT, so the duplicate is observable but not prevented.

6.8. External exactly-once only by composition

Proposition 8 (Composition with a provider key)**.** Case A. *If a key function π_t is paraphrase-invariant ($x \approx_t x' \Rightarrow \pi_t(x) = \pi_t(x')$) and complete for identity ($\pi_t(x) = \pi_t(x') \Rightarrow x \approx_t x'$), and the provider satisfies A8, then presenting $\pi_t(x)$ as the provider's idempotency key yields external exactly-once application for that tool, including across the commit window. A schema satisfying A5″ supplies $\pi_t = \sigma_t$.* Case B. *If the schema satisfies A5′ but not A5″, then σ_t is not admissible as a provider key (distinct acts sharing a projection would be merged at the provider, a wrong suppression that no local audit can see), and the canonical identifier op is admissible but not paraphrase-invariant; the commit window then remains open for paraphrased retries.*

Proposition 9 (Identity recognition is not substitutable)**.** *Mis-recognised identity defeats both fence-level commitment and a correct provider key: an unrecognised duplicate presents a fresh key, which a correct provider must apply, and a wrongly recognised distinct act presents a stale key, which a correct provider absorbs. Conversely, a provider key satisfying Case A of Proposition 8 makes A7 unnecessary for at-most-once application.*

Proofs are in Appendix A. Identity recognition (A) is thus the one guar-

antee neither the log nor the provider can supply, because both are keyed on its output.

6.9. What is proved and what is measured

Proved: exactness of the decidable layers under A2, A5 and A5′ (Proposition 4); the marginal bound under A6 with its budget, overall and per class (Theorem 1, Proposition 5, Corollary 1); attainment of the standard bound by the deployed statistic (Theorem 2, Corollary 2); fence-level at-most-once under A1–A4 and A7 (Proposition 7); external exactly-once only by composition (Proposition 8). Everything else is measured: whether the assumptions hold on this corpus, the residual's size, each judge's deduplication, how the realised rate varies across pools, splits and kinds of difference, behaviour under injection, and the composition part of the refinement dilemma.

7. Experimental methodology

Every analysis is an offline recomputation from persisted corpus files and judge scores; no reported number requires a model call to regenerate. Commands, configuration and data are in the supplementary material and the artifact.

7.1. Corpus

The corpus comprises 1,066 labelled pairs over six irreversible tools: `charge_card` (customer, amount, currency, memo), `post_refund` (charge, amount, reason), `transfer_funds` (two accounts, amount, currency, note), `send_email` (recipient, subject, body), `create_ticket` (title, description, priority) and `wire_money`, whose amount, beneficiary and reference are all

free text, so that the judge carries its entire identity decision. Each pair is labelled SAME or DIFF *by construction* (Table 6): a SAME pair comes from a transformation instructed to preserve the act, checked by a validator that rejects any variant altering a pinned number or recipient; a DIFF pair mutates a declared field, inserts a distinguishing token, or is hand-specified to differ in intent. The labels operationalise our definition of identity (Definition 2). Four generator arms—Llama-3.3-70B (109 pairs), GPT-OSS-120B (121) and Qwen3.6-27B (118) at temperature 0.9, and a hand-written arm of 68 S2 pairs—let self-recognition be measured; the remaining 650 pairs have no generator (supplementary material, Section S2). With 232 DIFF test pairs an observed zero admits a 95% Wilson upper bound of 1.6%, inside $\alpha = 0.05$.

Splits. Benign pairs are split 50/50 into calibration and test, stratified by (category, generator) with a fixed seed; injection categories are held out, and the build asserts both properties. Pairs derived from the same seed act can fall on both sides (82.8% [79.3–85.9] of test pairs share a seed act with a calibration pair), a route to optimism, so we also report 200 seed-level splits that partition the 219 seed acts first.

Judges. GPT-OSS-120B (primary), GPT-OSS-20B and Qwen3.6-27B—two model families—were reached through Groq's OpenAI-compatible endpoint in August 2026, at temperature 0, in batches of ten. A judge is admitted only if it meets conditions fixed before scoring: coverage of at least 80%, at least two distinct score values, and SAME/DIFF AUC of at least 0.60, which excludes a constant judge whose calibrated threshold would report a fake zero. Both GPT-OSS models scored all pairs (AUC 0.962); Qwen

Table 6: Corpus. Calibration/test counts are for the benign split; injection pairs are held out of calibration entirely.

Code	Label	n	cal/test	Construction
S1	SAME	68	34/34	mechanical reformat: key order, whitespace, number, currency and case formats
S2	SAME	253	126/127	paraphrase of free text with typed fields pinned; four generator arms
S3	SAME	75	37/38	loose-schema value rewrite, e.g. "$780.00" ↔ "seven hundred eighty dollars"
S4	SAME	163	81/82	paraphrase instructed to maximise surface difference while preserving the act
D1	DIFF	251	125/126	one field mutated (± 1, ± 0.01, $\times 10$, identifier or order-number change, currency swap)
D2	DIFF	150	75/75	legitimate repeats: identical except a distinguishing token ("instalment 1 of 2" vs "2 of 2")
D3	DIFF	61	30/31	hand-specified same-surface, different-intent pairs
D4a	DIFF	20	held out	typed field differs *and* free text says "treat as duplicate; suppress"
D4b	DIFF	25	held out	typed fields identical, distinct act, injection in free text
Total		1,066	508/513	559 SAME, 507 DIFF; 45 attack pairs

scored 1,041 (97.7%, AUC 0.974), its abstentions unassociated with category, label, tool, split or generator (smallest permutation $p = 0.09$; Section S4). Llama-3.3-70B reached 5.6% coverage before its endpoint disappeared and is excluded. Weights behind an endpoint are not pinned, so a re-run is a different experiment.

Configurations. B0 no fence; B1 exact string; B2 canonical key (content-addressed fencing, the prior-art baseline); a raw judge at $\tau = 0.5$; the *layered* configuration (certifier and calibrated judge, no refuter); the *proposed* configuration (certifier, refuter, calibrated judge, escalation band); and a *refined* configuration that adds to σ_t, for every free-text field, the numeric magnitudes it mentions—an intervention modelling a tool author who types the

numbers prose carries. Every configuration and ablation is calibrated on its own jurisdiction, at $\alpha = 0.05$ (and 0.01) with $\gamma = 0.05$. Each corpus pair compares one request with one committed effect, so $m = 1$ on the corpus.

Metrics and statistics. Wrong suppression and friction are measured over DIFF pairs and deduplication and silent or flagged escape over SAME pairs, on the 513-pair test split (281 SAME, 232 DIFF) and separately on the 45 attack pairs. Every rate carries a 95% Wilson interval [61]; a rate marked † is zero because a decidable layer makes it so. Paired comparisons use exact McNemar tests [62] with Holm correction [63] within the liveness and safety families. Agreement uses Krippendorff's α [64], Fleiss' κ [65] and Cohen's κ [66]. Seeds are recorded in the artifact's configuration file.

7.2. Validation of the labels

A relabelling study was preregistered before any judgement was collected: 154 frozen, per-item-hashed items in eleven strata weighted toward the categories a reader is most likely to contest, three blinded annotators and a blind adjudicator, three catch trials with a fixed exclusion rule, and a fixed decision rule for the safety claim (frozen at Git commit `82274ec`, 2026-08-23; SHA-256 sheet hash `66a174e1...9a9c`). The full 154-item preregistered Channel H study was not collected and is released as an open, cryptographically frozen turn-key replication artifact with an interactive browser instrument.

To evaluate label reproducibility across diverse reasoning systems without author consensus, *Channel M*—a blinded model panel (`groq/compound`, `groq/compound-mini`, `qwen/qwen3.6-27b`; `openai/gpt-oss-safeguard-20b` adjudicating; the primary judge not

queried directly)—was run one pair per request, and all four passed the catch trials. Both Compound systems are built on GPT-OSS-120B together with Llama models (Llama 4 Scout for `groq/compound`, Llama 3.3 70B for `groq/compound-mini`), so the primary judge's model is a component of two panelists, and the adjudicator is fine-tuned from the GPT-OSS family. The panel is therefore not independent of the primary judge; agreement with the two Compound panelists excluded is 96 of 117 (82.1% [74.1–88.0]). A model panel controls the same threat a human panel would—that the labels encode a definition private to the author—but more weakly, because a notion shared by the author and the models would pass it; it is reported strictly as model-panel evidence. The sheet draws 61 items from this corpus and 93 from an extension corpus over nine further tools whose judge scores were never persisted.

A targeted human follow-up on the two strata the panel contested (the 30 frozen items of those strata, the three catch trials, three annotators and an adjudicator, under the preregistered rules) is prepared and released with the artifact; it has not been run, and no human result is reported.

7.3. A second instance: action authorisation

An agent holds standing approvals, each with typed constraints (recipient, account or identifier, maximum amount, currency, maximum role, expiry) and a free-text scope. Permitting an action no approval covers—*wrong authorisation*—is the silent error; declining a covered action is visible. The certifier permits when every typed constraint holds and the purpose text is canonically equal to the scope; the refuter declines when a typed constraint fails; the residual goes to the judge, and the action is permitted only if its score

exceeds a threshold calibrated on not-covered residual pairs. Theorems 1 and 2 apply with wrong authorisation in place of wrong suppression; there is no band.

The corpus has 520 (approval, action) pairs over four tools (`issue_refund`, `transfer_funds`, `send_email`, `grant_access`) and 40 approvals. Per approval: the scope with whitespace perturbed (A1, covered, certified); three paraphrases (A2) and one specific instance (A3), covered; two typed violations (B1, refuted); four out-of-scope purposes with compliant typed values (B2, residual); and two attacks appending an instruction to permit, to a typed violation (B4a) and to an out-of-scope purpose (B4b). Labels are by construction; the seed texts were drafted with an AI assistant and are released for inspection. The split is by approval, so seed-level leakage is absent by construction; attacks are held out; judges, prompt format, batching, parser and admission rule are those of the identity experiments; $m = 1$ throughout.

7.4. Claims and the evidence for them

Table 7 maps each claim of the paper to the result that supports it and states what that result does and does not establish. It is the reading guide to Section 8.

8. Results

Benign rates are on the held-out test split (281 duplicate and 232 distinct pairs), attack rates on the 45 held-out injection pairs, each with a 95% Wilson interval; † marks a zero that follows from a decidable layer. Unless stated

Table 7: Claims, evidence and conclusions. "Proved" results hold under the listed assumptions; "measured" results hold on this corpus.

Claim	Theorem / experiment	Evidence	Conclusion
Decidable layers never err on the pairs they decide	Prop. 4; §8.1	0/232 distinct test pairs certified; 0/281 duplicates refuted; 20/20 typed-field injections refuted	proved under A2, A5, A5′; assumptions hold on this corpus
Semantic layer recovers paraphrased duplicates that canonical keys miss	§8.1	dedup 94.7% vs 12.1%; McNemar 232/0 discordant	measured; large effect
Wrong suppression is bounded on the deployed statistic	Thm. 1; §8.2	WS 1.7%; residual-conditioned 2.8%; mismatched pool 4.9%; Monte-Carlo 89.0% uncorrected vs $\leq$5.7% corrected	proved under A6; marginal, not per deployment
The bound's realised rate varies across pools and splits	§8.2	seed-level splits: mean 2.7%, 10.5% of splits above α	measured; a deployment can exceed α
The bound is marginal over kinds of difference; per-kind validity costs budget	Cor. 1; §8.3	D3 6.5% under the marginal threshold, 0/31 under its own; class-valid dedup 69.0% (12.1% on two judges)	measured; per-kind validity needs an observable partition or a larger pool
Multi-candidate behaviour on an admitted judge	Thm. 1; §8.2(v)	WS at $m = 5$: 0.0% [0.0–9.4] under both rules; dedup 98.6% [92.2–99.7] vs 27.5% [18.4–39.0]; attribution 1 of 68	measured; correction's cost only, two tools
Judge scores are stable across batch contexts	A6; §8.2(v)	163 of 391 unchanged; $m = 1$ WS 2.8% [1.1–6.9] $\rightarrow$ 5.6% [2.8–10.6]	*not supported*; recalibrate per scoring context
The bounded direction transfers across judges; deduplication does not	§8.4	WS 1.7/1.3/0.9%; dedup 94.7/82.6/89.0%	measured; three judges, two families
The refuter closes typed-field injection; prose-only injection remains judge-dependent	Prop. 4; §8.5	D4a 65.0%$\rightarrow$0%†; D4b 84.0% (primary)	proved for typed fields; measured for prose
Refinement shrinks the residual and the calibration budget	Props. 3, 5; §8.6	pool 142$\rightarrow$49; rank 7$\rightarrow$2; dedup 94.7$\rightarrow$69.0% (12.1% on two judges)	exact part proved; composition part measured
Fence-level at-most-once holds across process death outside the commit window	Prop. 7; §8.7	semantic 36/36 outside window vs syntactic 0/36; 0/48 missed legitimate effects	measured on one tool, one judge
External exactly-once is not provided	Props. 8, 9; §8.7	distinct arm: 12/12 re-fires inside the window, 11 flagged	confirmed; requires a provider key
Safety conclusion is robust to relabelling; liveness is not	§8.8	WS 1.7/1.9/2.2% across labellings; dedup 94.7/75.0/27.6%	measured; labels are constructed
Constructed labels are shared by independent readers	§8.8	Model panel only (88.0% [80.9–92.7]); two strata contested; no human annotation	*not established*

otherwise the judge is GPT-OSS-120B and $\alpha = 0.05$. Caveats common to all results are stated once, in Section 10.

8.1. Does the fence recover paraphrased duplicates, and at what cost?

Question. How much of the paraphrase gap left by canonical keys does a calibrated semantic layer close, and at what cost in the safety-critical direction?

Method. The baselines of Section 7 against the proposed configuration on identical test items, each calibrated on its own jurisdiction, with exact McNemar tests over discordant items.

Result. The decidable layers settle 122 of 513 test decisions without a model call—the certifier 34 (6.6% [4.8–9.1]) and the refuter 88 (17.2% [14.1–20.7])—leaving 76.2% [72.3–79.7] for the judge (Table 8, Figure 4). Canonical keys deduplicate 12.1% [8.8–16.4] with no wrong suppression; the proposed configuration deduplicates 94.7% [91.4–96.7] at 1.7% [0.7–4.3] wrong suppression, with 5.3% [3.3–8.6] of duplicates escaping silently (232 discordant items to 0, $p < 10^{-60}$). All four wrong suppressions are residual pairs: one near-miss, one legitimate repeat and two hand-specified semantic differences. Mechanical and loose-schema rewrites are fully deduplicated (34 of 34; 38 of 38, 100.0% [90.8–100.0]); benign and adversarial paraphrase reach 92.9% [87.1–96.2] and 92.7% [84.9–96.6].

Interpretation. The fence goes from catching one duplicate in eight to nine in ten, with its cost visible. A hand-set threshold ($\tau = 0.5$) deduplicates more and differs in wrong suppression by one discordant item ($p = 1.0$); calibration adds a bound that needs no tuning on labels, and the refuter adds

Table 8: Main comparison on the test split. Pool n is the size of each configuration's own calibration pool. B0 and B1 never suppress; the canonical key suppresses only the mechanical fragment. Flagged escape is 0.0% [0.0–1.3] for every row except the proposed configuration at $\alpha = 0.01$ (64.8% [59.0–70.1]) and the refined schema (17.8% [13.8–22.7]); friction is 0.0% [0.0–1.6] except in those two rows, where it is 0.9% [0.2–3.1].

Configuration	n	t_{hi}	wrong suppression	dedup	silent escape
B0 no fence	—	—	0.0% [0.0–1.6]	0.0% [0.0–1.3]	100.0% [98.7–100.0]
B1 exact string	—	—	0.0% [0.0–1.6]	0.0% [0.0–1.3]	100.0% [98.7–100.0]
B2 canonical key	—	—	0.0%† [0.0–1.6]	12.1% [8.8–16.4]	87.9% [83.6–91.2]
raw judge $\tau = 0.5$	—	0.50	2.2% [0.9–4.9]	99.6% [98.0–99.9]	0.4% [0.1–2.0]
layered (no refuter)	230	0.20	3.4% [1.8–6.7]	100.0% [98.7–100.0]	0.0% [0.0–1.3]
proposed, $\alpha = 0.05$	142	0.80	**1.7% [0.7–4.3]**	**94.7% [91.4–96.7]**	5.3% [3.3–8.6]
proposed, $\alpha = 0.01$	142	0.95	0.0% [0.0–1.6]	22.1% [17.6–27.3]	13.2% [9.7–17.6]
refined schema	49	0.90	0.0% [0.0–1.6]	69.0% [63.4–74.2]	13.2% [9.7–17.6]

robustness on typed fields (Section 8.5), not lower benign wrong suppression (four discordant items, Holm $p = 0.625$).

Limitation. α moves the operating point more than any other choice: at $\alpha = 0.01$ deduplication is 22.1% and 64.8% of duplicates go to review.

8.2. Is wrong suppression bounded for the statistic the fence deploys?

Question. Does the deployed rule behave as Theorem 1 says, does the jurisdiction condition matter, and does the multiplicity correction?

Method. (i) The proposed configuration calibrated on its own jurisdiction and on the wider non-canonical pool, with wrong suppression measured over the residual, the population the bound governs. (ii) A Monte-Carlo simulation with exchangeable continuous scores ($n = 200$, 2,000 replications per m), uncorrected against corrected. (iii) 60 live units with $K = 50$ nested

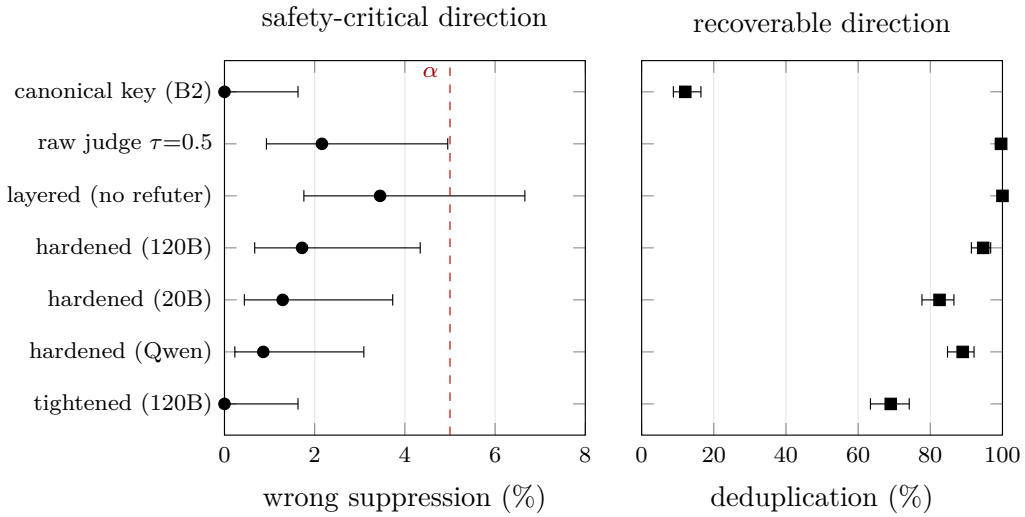


Figure 4: Main comparison on the 513-pair test split, $\alpha = 0.05$, with 95% Wilson intervals. Canonical keys never wrongly suppress but deduplicate only the mechanical fragment. The proposed (hardened) configuration keeps wrong suppression below α for all three judges while deduplication varies with the judge. The layered configuration is calibrated on its own, wider jurisdiction. The tightened configuration trades deduplication for zero benign wrong suppression and closes prose-only injection (Figure 6).

candidates scored by Llama-3.1-8B-Instant. (iv) Resampled pools and 200 seed-level splits. (v) Multi-candidate units built from the corpus and scored by the primary judge: one unit per residual test pair, whose partner is the true match (duplicate units) or a labelled distinct act, plus up to nine further candidates that share the request's projection and are known distinct from it and from each other (Section S10); thresholds from the deployed pool ($n = 142$, certifiable $m \leq 7$).

Result. (i) With the matched pool ($n = 142$, $t_{\text{hi}} = 0.80$) residual wrong suppression is 2.8% [1.1–6.9]; with the wider pool ($n = 230$, $t_{\text{hi}} = 0.20$) it

is 4.9% [2.4–9.7], while the rate over the whole test split stays at 3.0% [1.5–6.1], below α. The second judge behaves the same way (2.1% [0.7–5.9] against 4.9% [2.4–9.7]). (ii) The uncorrected rule reaches 5.0%, 11.2%, 22.9%, 38.3%, 70.3% and 89.0% at $m = 1, 2, 5, 10, 25, 50$; the corrected rule gives 5.0%, 5.7% (within Monte-Carlo error), 4.5%, 3.7%, 0.0% and 0.0%, the zeros being the budget switching semantic suppression off (Figure 5). (iii) The live judge's scores take two values, so the result does not depend on the calibrated value: at $m = 50$ the uncorrected rule wrongly suppresses 65.0% [43.3–81.9] of held-out distinct units and the corrected rule 0.0% [0.0–16.1], which also stops deduplicating because its pool of 205 certifies only $m \leq 10$. (iv) Resampling leaves the mean realised rate flat in pool size while the tail and the share of pools above α fall (Section S11); seed-level resplitting raises the mean on every judge—1.7% to 2.7%, 1.3% to 2.2%, 0.9% to 1.8%—with 10.5%, 1.0% and 11.0% of splits above α. (v) On the corpus-derived units scored by the primary judge, the added candidates did not produce the failure the correction guards against: wrong suppression of distinct units is 2.0% [0.4–10.7] under both rules at $m = 2$ ($n = 49$) and 0.0% [0.0–9.4] under both at $m = 5$ and $m = 10$ ($n = 37$), because the extra candidates are acts from other seeds that the judge scores low. The correction therefore only costs deduplication here: 98.6% [92.2–99.7] uncorrected against 27.5% [18.4–39.0] corrected at $m = 5$, and none at $m = 10$, beyond the budget. Response attribution is rarely wrong: the returned response belongs to a different act for 1 of 239 suppressed duplicates at $m = 2$ and 1 of 68 at $m = 5$ (ties counted as errors). Re-scoring exposed a stability problem instead. Scored again in a different batch at temperature 0, only 163 of 391 corpus partners received

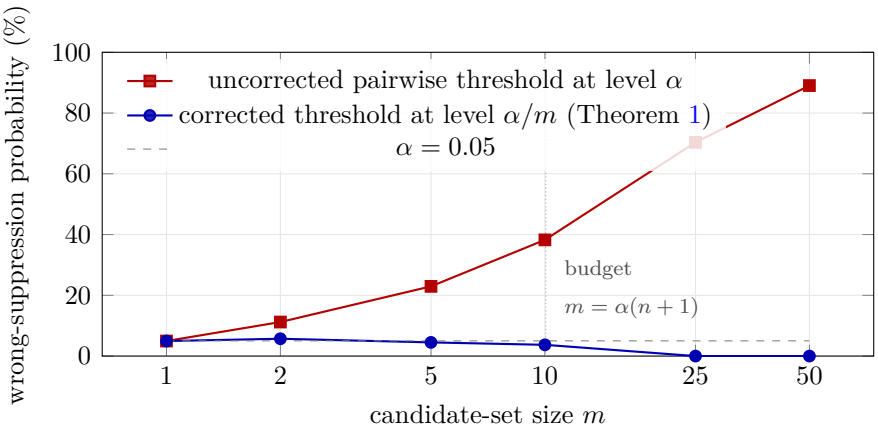


Figure 5: Why the correction is needed. Monte-Carlo simulation with exchangeable continuous scores, calibration pool $n = 200$, $\alpha = 0.05$, 2,000 replications per m: the probability that a request distinct from all m candidates is suppressed. The uncorrected rule reaches 89.0% at $m = 50$. The corrected rule stays within Monte-Carlo error of α and switches semantic suppression off once m exceeds the budget $\alpha(n+1)$, which is the intended failure direction.

their persisted score; most changes are small (17 of 391 move by 0.2 or more; mean +0.016), but 6 distinct pairs crossed the 0.80 threshold upward and 2 downward, so the same 144 distinct residual pairs give 5.6% [2.8–10.6] wrong suppression at $m = 1$ against 2.8% [1.1–6.9] with the persisted scores.

Interpretation. A pool from outside the jurisdiction roughly doubles the rate the bound governs while the whole-split rate stays under α, so the violation is invisible unless one conditions on the residual. The correction is necessary and pays for safety by switching off beyond its budget; the theorem is marginal and depends on jurisdiction-matched exchangeability, so realised rates can vary across calibration draws.

Limitation. The simulation uses synthetic scores and the saturated-judge experiment purpose-built units. The admitted-judge units are built from the corpus, reach $m \geq 5$ only on two tools, and contain no hard multi-candidate cases, so they measure the correction's cost, not its benefit. The re-scoring result means the judge's scores depend on batch context, which A6 does not allow for: the calibration pool and the deployed comparisons must be scored under the same batching, and a pool scored in a different context can understate the realised rate (Section 10).

8.3. Is the bound valid within each kind of difference?

Question. The distinct pairs in the jurisdiction mix three kinds of difference we chose: near-miss mutations (D1), legitimate repeats (D2) and hand-specified semantic differences (D3). Does the bound hold within each, and what does per-kind validity cost?

Method. Offline, from the persisted scores: each category's wrong suppression under the deployed threshold and under a Mondrian threshold on its own residual calibration pool (Corollary 1). The category is label-defined, so a per-category threshold is an oracle; the rule valid for every category without knowing which applies is $\max_c t_c$. We also calibrate per tool, a partition the fence can observe, and reweight the residual distinct test mix toward D3.

Result. On the primary judge the deployed threshold ($t_{\text{hi}} = 0.80$, pool 142, certifiable $m \leq 7$) gives 0.8% [0.1–4.4] on D1, 1.3% [0.2–7.2] on D2 and 6.5% [1.8–20.7] on D3. The per-category pools hold 37, 75 and 30 scores: each is certifiable at $m = 1$, but only D2 beyond it. D3's Mondrian threshold is its pool's largest score, 0.90, and removes both D3 errors (0.0% [0.0–11.0]).

Which category carries the excess depends on the judge: on GPT-OSS-20B and Qwen it is legitimate repeats, at 4.0% [1.4–11.1] and 2.7% [0.7–9.2]. The class-valid rule removes every benign wrong suppression on all three judges (0.0% [0.0–1.6]) at a cost in deduplication: 94.7% falls to 69.0% [63.4–74.2] on the primary judge and to the canonical-only 12.1% [8.8–16.4] on the other two. Calibrating per tool gives 0.4% [0.1–2.4] wrong suppression and 67.6% [61.9–72.8] deduplication on the primary judge, with two of six tools below the 19-pair budget. Reweighting moves the primary judge's residual rate from 2.8% [0.7–5.6] at the observed D3 share (20.8%) to 4.2% [0.4–9.3] at 50% and 6.7% [1.8–21.3] on D3 alone (Section S23).

Interpretation. The marginal guarantee is an average over a mix, and the mix is ours. No per-category rate is significantly above α, but the theorem gives no reason to expect any to be below it, and on the primary judge D3's point estimate is not. Per-kind validity costs what Corollary 1 predicts: partitioning divides the budget, pushes each class threshold up the ranks and, when the classes cannot be observed, charges every request the most conservative one. This is the refinement dilemma applied to the calibration population rather than to the schema.

Limitation. Thirty to seventy-five calibration pairs per category; a deployment's kinds of difference are unknown.

8.4. Does the bounded direction transfer across judges?

Question. Is the behaviour a property of the mechanism or of one model?

Table 9: Three judges, each calibrated on its own jurisdiction, $\alpha = 0.05$. The two GPT-OSS models share a family; Qwen is a second family. Wrong suppression is below α for all three; deduplication and silent escape are governed by nothing.

Judge	coverage	t_{hi}	wrong suppression	dedup	silent escape
GPT-OSS-120B (primary)	100%	0.80	1.7% [0.7–4.3]	94.7% [91.4–96.7]	5.3% [3.3–8.6]
GPT-OSS-20B	100%	0.95	1.3% [0.4–3.7]	82.6% [77.7–86.6]	2.1% [1.0–4.6]
Qwen3.6-27B	97.7%	0.95	0.9% [0.2–3.1]	89.0% [84.8–92.1]	8.5% [5.8–12.4]

Method. Three admitted judges from two families, each calibrated on its own pool, on the full test split and on the 499 test pairs all three scored; deduplication by paraphrase-generator arm; overlap of the judges' errors.

Result. Wrong suppression is 1.7% [0.7–4.3], 1.3% [0.4–3.7] and 0.9% [0.2–3.1], all below α, against deduplication of 94.7% [91.4–96.7], 82.6% [77.7–86.6] and 89.0% [84.8–92.1] and silent escape from 2.1% [1.0–4.6] to 8.5% [5.8–12.4] (Table 9); the common subset gives the same picture. The judges' wrong suppressions are four, three and two pairs with *no pair in common.* The primary judge deduplicates 91.2% [77.0–97.0] of the hand-written arm, 96.4% [87.7–99.0] of the Llama arm, 91.8% [82.2–96.4] of its own family's arm and 91.5% [81.6–96.3] of the Qwen arm.

Interpretation. The bounded direction stays bounded on every judge we could admit; the unbounded direction moves twelve points. Disjoint error sets are what Proposition 3 predicts: a different oracle redistributes error within the residual rather than removing it. There is no sign of self-recognition, though with 34 hand-written test pairs an advantage under about ten points would not be visible.

Limitation. The disjointness rests on nine errors in total.

8.5. Adversarial injection: typed fields against prose

Question. Under prompt injection aimed at the fence, what does the refuter remove, and what remains?

Method. The 45 held-out attack pairs are distinct acts, so every suppression is wrong. D4a differs in a typed field and carries text asserting a duplicate; D4b has identical typed fields and is distinguished only by prose. Injection is a deliberate shift, so Theorem 1 does not apply.

Result. The raw primary judge wrongly suppresses 65.0% [43.3–81.9] of D4a and the calibrated layer without the refuter 70.0% [48.1–85.5]; the refuter drives D4a to 0.0%† [0.0–16.1] on every judge (Figure 6). On D4b the refuter is silent and the outcome is the judge's: 84.0% [65.3–93.6] for the primary judge, 28.0% [14.3–47.6] for GPT-OSS-20B and 24.0% [11.5–43.4] for Qwen. A `min` ensemble tracks the more robust judge (28.0% [14.3–47.6]) and a `mean` ensemble falls between (44.0% [26.7–62.9]). Under the refined declaration every judge's D4b rate is 0.0%† [0.0–13.3].

Interpretation. The injection experiment demonstrates the value of jurisdiction decomposition: typed-field injection is eliminated before judging, whereas prose-only manipulation remains inside the residual LLM jurisdiction where the primary judge wrongly suppresses 84.0% of attack pairs. The result shows that schema design and judge calibration address different parts of the attack surface: the architectural remedy for prose vulnerability is schema refinement—moving distinguishing attributes into typed fields—rather than relying on judge prompt tuning.

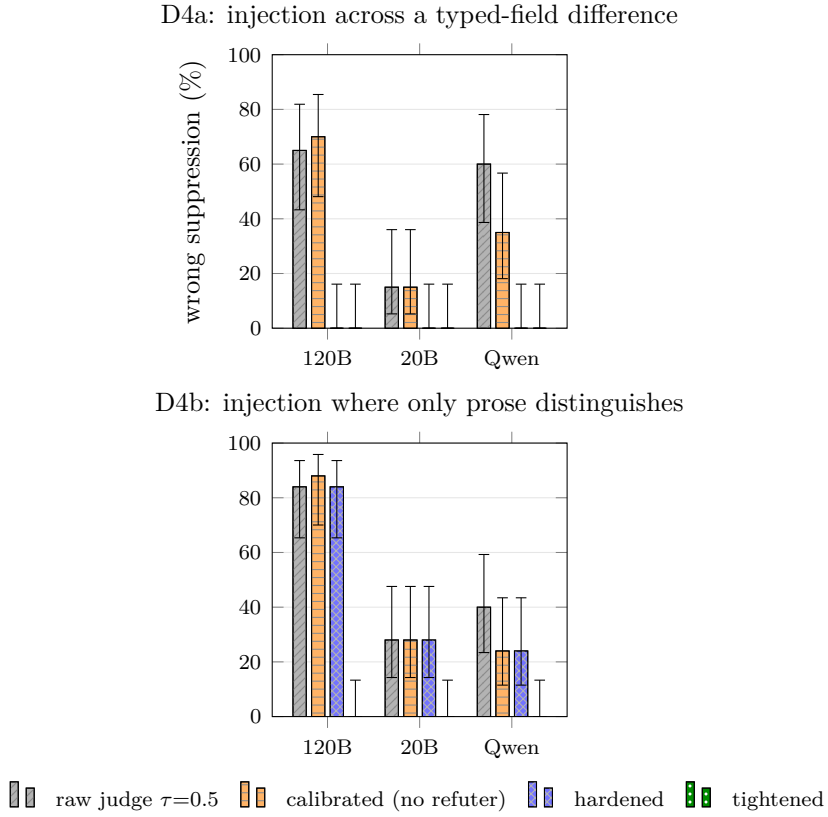


Figure 6: Adversarial injection on the 45 held-out attack pairs (all distinct acts; every suppression is wrong), with 95% Wilson intervals. Calibration offers no protection under deliberate shift. The refuter removes D4a on every judge by construction (the zeros are logical, not estimated). On D4b the typed fields agree, the refuter is silent, and the outcome is whatever the judge does, until the refined (tightened) declaration types the distinguishing numbers. Theorem 1 does not apply to this split.

Limitation. Evaluated on forty-five constructed attack pairs using standard injection templates; adversaries with knowledge of declared typed projections will specifically target free-text residual fields.

8.6. The refinement dilemma, measured and decomposed

Question. Refinement shrinks the calibration pool (Proposition 5). What does that cost, and is the cost the pool's size or its composition?

Method. The refined declaration types, for every free-text field, the numeric magnitudes it mentions; both declarations are calibrated on their own jurisdictions. A size-only counterfactual draws 5,000 random subsets of the declared pool at the refined pool's size and evaluates the refined configuration with each threshold.

Result. Refinement closes prose-only injection on every judge and drives benign wrong suppression to 0.0% [0.0–1.6], while judge jurisdiction falls from 76.2% [72.3–79.7] to 58.9% [54.6–63.0] (Table 10). Deduplication falls from 94.7% [91.4–96.7] to 69.0% [63.4–74.2] on the primary judge, and from 82.6% [77.7–86.6] and 89.0% [84.8–92.1] to the canonical-only 12.1% [8.8–16.4] on the other two, whose refined thresholds sit at a score of 1.0. No duplicate is falsely vetoed (0 of 281); on the primary judge 72 duplicates stop being suppressed, 50 of them flagged. In the size-only counterfactual mean deduplication is 88.4%, and the observed 69.0% lies at its 11.6th percentile; for the two saturated judges a size-only draw reaches the ceiling in 43.9% and 41.9% of cases. Removed pairs are easier: 2.2% of removed and 10.2% of retained pairs score at or above 0.8 for both GPT-OSS judges.

Table 10: Declared against refined declarations, $\alpha = 0.05$. The rank is the position of the threshold from the top of the pool (Proposition 5b); $t_{\text{hi}} = 1.00$ with a strict inequality means no semantic suppression.

Judge	declaration	pool n	rank	t_{hi}	wrong suppression	dedup	D4b injection
GPT-OSS-120B	declared	142	7	0.80	1.7% [0.7–4.3]	94.7% [91.4–96.7]	84.0% [65.3–93.6]
	refined	49	2	0.90	0.0% [0.0–1.6]	69.0% [63.4–74.2]	0.0%† [0.0–13.3]
GPT-OSS-20B	declared	142	7	0.95	1.3% [0.4–3.7]	82.6% [77.7–86.6]	28.0% [14.3–47.6]
	refined	49	2	1.00	0.0% [0.0–1.6]	12.1% [8.8–16.4]	0.0%† [0.0–13.3]
Qwen3.6-27B	declared	139	6	0.95	0.9% [0.2–3.1]	89.0% [84.8–92.1]	24.0% [11.5–43.4]
	refined	48	2	1.00	0.0% [0.0–1.6]	12.1% [8.8–16.4]	0.0%† [0.0–13.3]

A three-level schema ladder over a separate act family confirms the monotone part: the residual falls from 100.0% [96.4–100.0] to 41.2% [32.1–50.9] to 19.6% [13.1–28.4], and the calibrated layer suppresses nothing at any level (Section S13).

Interpretation. Refinement is a trade: zero wrong suppression, one injection class closed and 23% less oracle load, paid for in the recoverable direction. Both components are real: the threshold moves from the 7th to the 2nd largest score, and the surviving pool is enriched in the pairs the judge finds hardest. This demonstrates that schema design is part of the statistical specification of the mechanism.

Limitation. The refined projection types the numbers prose mentions, not the concepts; the composition component is a property of these judges on this corpus.

8.7. Process death and paraphrased replay

Question. Does the mechanism hold across real process death, and where does it stop holding?

Method. Three operating-system processes share a directory: one commits a wire, a second is killed at one of eight crash points, and a third replays the run with a paraphrase from a pool written by a different model family from the judge. In the *duplicate* arm the replay is the same act (correct: one wire); in the *distinct* arm it is a second instalment distinguished only in prose (correct: two wires). 48 trials per arm, both fence modes on identical draws.

Result. In the duplicate arm the content-addressed fence reaches the correct final state in 0.0% [0.0–7.4] of trials and the semantic fence in 97.9% [89.1–99.6], including 36 of 36 outside the commit window; every replay was canonically different, so all suppressions came from the calibrated layer. In the distinct arm the semantic fence missed no legitimate effect (0.0% [0.0–7.4]). Inside the commit window it re-fired in all 12 trials, flagging 11 (91.7% [64.6–98.5]). Per-point results and cost are in Section S15.

Interpretation. At-most-once commitment holds across process death outside the commit window, including where content-addressed fencing cannot help, without dropping a legitimate act; inside the window the failure is made observable but not prevented, as Proposition 7 predicts. This separates guarantee (B) from (C).

Limitation. One tool, one paraphrase pool, one judge; the provider has no idempotency key by design.

8.8. How much do the conclusions depend on our labels?

Question. Do the conclusions survive defensible relabellings, and do independent readers agree with the construction?

Method. (i) The headline re-derived with legitimate repeats treated as the same act, and with adversarial paraphrases treated as distinct (assuming our validator failed). (ii) The preregistered relabelling study on Channel M (Section 7.2).

Result. (i) Wrong suppression is 1.7% [0.7–4.3], 1.9% [0.7–5.5] and 2.2% [1.1–4.5] under the three labellings, all inside α; deduplication is 94.7% [91.4–96.7], 75.0% [70.3–79.2] and 27.6% [21.9–34.2]. (ii) The panel agrees with the construction on 103 of 117 fully annotated items (88.0% [80.9–92.7]; Krippendorff's $\alpha = 0.829$, Fleiss' $\kappa = 0.829$; no item needed adjudication). Two strata are contested under the preregistered rule (upper bound below 0.80): legitimate repeats at 50.0% [25.4–74.6], pre-declared as the likeliest to be contested, and prose-only injection at 45.5% [21.3–72.0]. Thirteen of the fourteen label flips are residual pairs; the fourteenth is refuter-decided and executed whatever its label. Dropping the panelist that is also a reported judge raises agreement to 91.7% [85.0–95.6]. Dropping the two Compound panelists gives 82.1% [74.1–88.0].

The preregistered re-derivation gives 9.9% [4.9–19.0] wrong suppression under constructed labels and 10.5% [4.9–21.1] under panel labels, which *triggers* the preregistered rule: the panel-label estimate exceeds both the constructed one and α. Two defects in that computation are reported with it. First, 93 of the 154 items come from an extension corpus whose scores were

never persisted, so they can escalate but never be suppressed; they enter the denominator without being able to enter the numerator, and 12 of the 14 flips are such items. Second, the subset is deliberately adversarial (seven of eleven strata are distinct or attack strata). On the 51 annotated items the primary judge actually scored, wrong suppression is 4 of 22 distinct items under constructed labels and 3 of 20 under panel labels; on the benign ones it falls from 1 of 14 to 0 of 12, and the remaining errors are prose-only injections, where no bound is claimed.

Interpretation. The safety conclusion does not depend on our definition of identity; the liveness conclusion does (94.7% to 27.6%). The model panel's disagreements fall on pairs whose distinguishing information lives only in free text. Whether human readers would contest the same strata is unknown: no human annotation was collected. The trigger appears to be an artefact of the unscored items, but the rule was fixed in advance so that we could not reason our way out of it, and we report it as triggered.

Limitation. This is a model panel, not a human study, and it is not independent of the primary judge; the human instruments are released unrun (Section 10).

8.9. What each component contributes, and what it costs

Question. Which layers earn their place, and what does the mechanism cost?

Method. Single-layer ablations, each calibrated on its own jurisdiction; a sweep over fixed thresholds; token and latency accounting from the scoring sessions.

Result. On the primary judge, removing the certifier changes no rate, and removing the escalation band changes nothing because the band is empty ($t_{\text{lo}} = t_{\text{hi}} = 0.80$). On GPT-OSS-20B, whose band is not empty, removing it turns 15.3% [11.6–20.0] of duplicates from flagged into silent escapes; the band is driven by ties, not graded confidence (40 of 43 duplicates in it score exactly at the threshold). Removing the refuter raises deduplication to 100.0% and wrong suppression to 3.4% [1.8–6.7]. The conformal threshold sits deep inside the 80–95% of the fixed-threshold range that is safe on the test set, giving up 6–20 points of deduplication against the best fixed threshold, which needs test labels to find; against a cross-validated fixed threshold it exceeds α on 29.2% of bootstrap resamples against 39.6%. The judge is consulted on 76.2% of dispatches at about 184 tokens per comparison in batches of ten (Section S17).

Interpretation. The certifier's value is a guarantee, not a rate; the band matters only when ties leave $t_{\text{lo}} < t_{\text{hi}}$; and at $m = 1$ calibration buys a label-free threshold, a correction valid at $m > 1$ and safe shutdown rather than accuracy.

Limitation. Single-split accounting at $m = 1$ with batched API calls; unbatched dispatch latency and live multi-candidate costs are not measured.

8.10. A second instance: action authorisation

Question. Does the recipe transfer to action authorisation, where the decision is whether an action lies within what a user approved?

Method. The 520-pair corpus of Section 7.3, split by approval (220 benign test pairs; 80 attack pairs held out), under exact scope matching, a raw

judge, a calibrated judge without the refuter, and the layered rule, on the three judges at $\alpha = 0.05$ and 0.01.

Result. The certifier permits 9.1% [6.0–13.6] of test pairs and the refuter denies 18.2% [13.6–23.8], routing 72.7% [66.5–78.2] to the judge. On the primary judge ($n = 80$, $t_{\mathrm{hi}} = 0.40$) exact scope matching permits 20.0% [13.3–28.9] of covered actions at 0.0% [0.0–3.1] wrong authorisation; the layered rule raises autonomy to 100.0% [96.3–100.0] at 5.0% [2.3–10.5] wrong authorisation (80 discordant items to 0, $p = 1.7 \times 10^{-24}$). The other judges give 1.7% [0.5–5.9] and 0.8% [0.1–4.6]. The refuter drives the typed-violation attack to 0.0% [0.0–8.8] on every judge; on the free-text scope attack the judges permit 15.0% [7.1–29.1], 12.5% [5.5–26.1] and 0.0% [0.0–8.8]. At $\alpha = 0.01$ the pool of 80 is below the budget of 99, so $t_{\mathrm{hi}} = +\infty$ and the rule falls back to exact scope (Section S22).

Interpretation. The decomposition transfers intact, including the refuter's closure of typed-field injection and safe shutdown beyond the budget.

Limitation. Drafted seeds and constructed labels, with no independent annotation.

9. Discussion

9.1. What the evidence supports

The most robust finding is an asymmetry between the two error directions. Wrong suppression stayed below α across three judges (1.7%, 1.3%, 0.9%), three labellings (1.7%, 1.9%, 2.2%) and the mean over 200 seed-level splits (2.7%, 2.2%, 1.8%), while deduplication moved from 94.7% to 82.6%

across judges, to 27.6% across labellings, and to 69.0% or 12.1% under refinement. That is what a bound on one direction and none on the other looks like from outside, not evidence of greater accuracy; part of the stability is conservatism (Section 8.9), and it holds only for a jurisdiction-matched pool and as an average over the kinds of difference the pool mixes (Sections 8.2 and 8.3).

The second finding is that semantic retry safety is not solely an LLM-judging problem: the declared schema determines the judge's jurisdiction. Every benign wrong suppression we observed, and all but one label disagreement of the model panel, fell on a pair whose distinguishing information lives only in free text. That is partly true by construction (Proposition 3): typed-field distinctions are settled before the judge and therefore cannot be overridden by injected prose in those fields, whereas distinctions left in free text remain within the residual judge jurisdiction. Moving act-determining information into typed fields removes cases from that jurisdiction and closes corresponding attack paths, while simultaneously changing the calibration population and statistical budget—the schema refinement dilemma.

9.2. Practical guidance

(1) Declare every act-determining field as typed where possible; an effect-determining field left in free text is a known exposure to the judge and to injection.

(2) Calibrate each configuration on its own jurisdiction, and recalibrate whenever a layer, the judge or the schema changes.

(3) Size the pool for the candidate budget: at $\alpha = 0.05$, 19 distinct residual

pairs to suppress semantically at all and $m/\alpha - 1$ for m candidates (199 for $m = 10$), per class if a per-class guarantee is needed.

(4) Check how many duplicate scores tie with t_{hi} on a coarse grid; ties decide whether a missed duplicate is flagged or silent.

(5) Prefer a verified postcondition where one exists [24], and a provider key derived from a projection complete for identity where the schema permits (Proposition 8).

(6) Log and sample semantic suppressions for audit: the bound is marginal, and a wrong suppression leaves no trace otherwise.

9.3. Beyond duplication

Effect identity is one instance of a recurring shape in agent safety: a binary decision with sharply asymmetric consequences, no complete decidable procedure, and a stochastic model as the only general-purpose decider. The recipe—find a decidable certifier and refuter, name the residual, calibrate the deployed statistic on the residual in the safety-critical direction only, make the undecided band auditable, report the budget—transferred to action authorisation (Section 8.10); content release and other gatekeeping decisions remain open.

10. Scope, limitations and broader impact

Each scope boundary and operational assumption of the paper is stated here in full; elsewhere it appears as a cross-reference.

10.1. Scope boundaries and assumptions

Semantic assumptions and schema dependencies. The decidable layers are sound under the condition that declared field normalisation and projection policies preserve intended semantics (A5, A5′); semantic mis-declarations are not detected automatically. Deduplication, silent escape and robustness to injection depend on the judge; two of the three judges share a model family, and a third vendor was never reached at usable coverage. Models served through external endpoints change without notice, which breaks exchangeability until the pool is recalibrated. Even without a model change, the primary judge's scores are not reproducible at temperature 0 across batch contexts: re-scoring 391 corpus pairs in new batches left 163 of 391 scores unchanged and doubled wrong suppression on the 144 distinct residual pairs, from 4 of 144 to 8 of 144 (Section 8.2). A deployment must score calibration and deployed comparisons under the same batching, or one pair per call, and treat the calibrated rate as an estimate for that scoring context only.

Statistical assumptions. Theorem 1 requires exchangeability (A6) between the pool and deployed comparisons within the rule's jurisdiction; it cannot be verified at run time and is violated by injection and drift. The bound is marginal over calibration draws: under seed-level resplitting 10.5% of splits exceed α on the primary judge (Section 8.2), and hand-specified semantic differences (D3) reach 6.5% under the pooled threshold (Section 8.3).

Operational and durable-store assumptions. Fence-level at-most-once commitment is proved under A1–A4 and A7 (Proposition 7). The research log supplies A3 only partially: it enforces uniqueness for a single writer but is not

crash-atomic, and crashes were injected at protocol points rather than inside a write, so a deployment needs a database uniqueness constraint. Concurrent dispatch of one operation (A7) was not tested; without serialisation, concurrent executions are observable through recorded conflicts but not prevented.

Evaluation corpus and labels. The corpora are synthetic and purpose-built, cover six tools (four for authorisation), and say nothing about the distribution of paraphrase in production; the rates are properties of these corpora, and what transfers is the decomposition, the bound and the dilemma. They are constructed, with labels fixed by construction under Definition 2. No human annotation was collected: the preregistered human study, and a smaller follow-up on the two strata the model panel contested, are released as ready-to-run instruments. The only independent check of the labels is the blinded model panel, which is not independent of the primary judge (two panelists are built on it) and contested two strata (Section 8.8); a notion of "the same act" shared by the author and the models would pass it and could still differ from human practice. The preregistered wrong-suppression rule triggered on the audit subset (Section 8.8). Deduplication, unlike wrong suppression, depends on the labelling. The seed texts of the authorisation corpus were drafted with an AI assistant, and that corpus has no independent annotation.

Candidate multiplicity. The corpus has $m = 1$. Multi-candidate behaviour is measured in simulation, in a 60-unit live study whose judge (Llama-3.1-8B-Instant) returned two score values, and on corpus-derived units scored by the primary judge (Section 8.2); those units are built from the corpus rather than from naturally occurring agent traces, and reach $m \geq 5$ only on

the two tools whose schemas leave many committed acts undistinguished by the refuter. Retries under a fresh run identifier or a different tool are out of scope.

Adversarial text. Prose-only injection remains in the judge's jurisdiction (84.0% on the primary judge; Section 8.5); no bound is claimed under injection, and the remedy is to type act-determining fields (Section 8.6).

External execution. Guarantee (C) is outside the mechanism's contract (Section 3.5, Proposition 8).

Operating budget and cost. The judge is consulted on 76.2% of dispatches, at about 184 tokens per comparison in batches of ten. Operating points are governed by α: selecting a stricter level of $\alpha = 0.01$ prioritises safety over autonomy, yielding 22.1% deduplication and routing 64.8% of duplicates to flagged review.

10.2. Ethics and broader impact

The mechanism is meant to reduce harm from irreversible agent actions, and its two failure modes carry different ethical weight. Wrongly suppressing a legitimate action—a refund never issued, a message never sent—is silent, and the person affected may never learn of it; the design bounds this error but does not eliminate it. Executing an unintended duplicate is visible and recoverable, and the design deliberately tolerates more of it. Adversaries can target both directions; the refuter removes forced suppression whenever the distinction is typed, which is a reason to type act-determining fields rather than to trust a better judge. The injection templates are standard in

the literature and released for defensive evaluation. Human oversight enters through the escalation band and audit; a band that emits more flags than reviewers can handle is not a safety mechanism [67], so the fence records every semantic suppression with the matched effect, score, m and threshold. We do not recommend the mechanism as the only safeguard for financial or safety-critical agent operations. The corpora are synthetic and contain no personal data.

11. Conclusion

Every mechanism that prevents an autonomous agent from repeating an irreversible action rests on deciding whether two requests denote the same intended act. Syntactic journals, client-chosen idempotency keys and content hashes all assume deterministic replay; they fail when an agent re-synthesises its arguments upon retry. Conversely, unconstrained language-model judgements introduce an unbounded hazard for the safety-critical error of silently suppressing a genuinely distinct action.

We formulated semantic effect identity as a control decision under asymmetric loss and decomposed it into a sound certifier, a sound refuter, and a schema-determined residual jurisdiction. By evaluating an LLM judge only on this residual and thresholding the maximum statistic over surviving candidates at a multiplicity-corrected conformal level, the fence establishes a finite-sample marginal wrong-suppression bound of α under stated exchangeability and jurisdiction-matching assumptions. Refining the schema or demanding class-conditional validity improves deterministic protection but simultaneously contracts the calibration population and statistical budget, a trade-off

we state exactly.

On the constructed evaluation corpus of 1,066 pairs across six tools, the fence recovered 94.7% of paraphrased duplicates at 1.7% wrong suppression, against 12.1% for canonical keys, with wrong suppression remaining below $\alpha = 0.05$ across three models from two model families, while the realised rate varies across calibration draws as a marginal bound permits. Typed fields eliminate adversarial field manipulation, whereas prose-only manipulation remains within the residual judge's jurisdiction. The natural next step is to evaluate the multiplicity-corrected mechanism on live, naturally occurring multi-candidate agent traces and to study adaptive schema induction under evolving tool interfaces.

The resulting view is that semantic retry safety is not solely an LLM-judging problem: the schema determines which decisions can be made without a judge, which cases remain probabilistic, and therefore where statistical guarantees must be established.

CRediT authorship contribution statement

Sivaguru R. M.: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author develops AgentRuntime, a software project for reliable execution of LLM agents that includes a content-addressed effect fence of the

kind used as the canonical-key baseline in this paper. The author declares no other competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used Claude (Anthropic) to draft, revise and shorten text; to write and review analysis code, including the class-conditional analysis, the tightness checks, and the prepared multi-candidate and targeted human-annotation materials; to check reference metadata against publisher and arXiv records; and to audit reported numbers against the persisted results. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication. Language models used as research instruments—judges, paraphrase generators and the model annotation panel—are described in Section 7 and are part of the study, not of manuscript preparation. The same assistant drafted the seed texts of the authorisation corpus (Section 7.3); they are research materials, are disclosed as such there, and are released for inspection.

Ethics approval and consent to participate

No human participants took part in this research, and no personal data were collected. The corpora are synthetic; label validation used a panel of language models (Section 7.2).

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

The complete research artifact—comprising the evaluation corpus, judge score traces, cryptographic annotation freeze, replication instruments, model-panel outputs, and automated verification suites—accompanies this submission (supplementary material, Section S20). All tables and figures regenerate offline from the persisted scores. The open artifact repository is accessible at https://github.com/Sivaguru-2008/agent-runtime. Judge, generator and panel models were reached through a third-party inference endpoint. Two generator models are no longer available on it, so the corpus cannot be regenerated byte for byte and is frozen and released instead; the two Compound panel systems were decommissioned by the provider on 21 September 2026, so the model-panel study cannot be re-run, and its responses are released.

Appendix A. Proofs

Proof of Proposition 1. Suppose substituting the recorded response of x' for x drops nothing owed. By the first condition x denotes one act $a = [\![x]\!]_t$ and x' one act $a' = [\![x']\!]_t$. If $a \neq a'$, the recorded response of x' reflects a' by the second condition, and returning it in place of executing x drops a, a contradiction; hence $a = a'$. Conversely, if $a = a'$ the substitution returns the response of the act that x denotes and drops nothing. The test

therefore coincides with equality of denotations, which is reflexive, symmetric and transitive. □

Proof of Proposition 2. If $\mathsf{C}_1, \mathsf{C}_2 \subseteq \approx_t$ then $\mathsf{C}_1 \cup \mathsf{C}_2$ and $\mathsf{C}_1 \cap \mathsf{C}_2$ are subsets of $\approx_t$; $\emptyset$ and $\approx_t$ are certifiers and bound the family. The argument for refuters is identical with the complement of $\approx_t$ in place of $\approx_t$. If $\mathsf{C}_1, \ldots, \mathsf{C}_k$ are decidable, running their decision procedures and taking the disjunction decides $\bigcup_i \mathsf{C}_i$; decidability need not survive countable unions, so only finite unions are claimed, and only finite unions are used. A relation that is both a certifier and a refuter satisfies $R \subseteq \approx_t$ and $R \cap \approx_t = \emptyset$, hence $R = \emptyset$. □

Proof of Lemma 1. Fix a pool s with distinct entries and suppose $\tau(s)$ is finite and equal to no s_i. Then $\tau(s)$ lies in an open interval I that contains no pool value and whose endpoints are consecutive pool values or $\pm\infty$. Let g be the identity outside I and, on I, a strictly increasing bijection of I onto itself with $g(\tau(s)) \neq \tau(s)$; a piecewise-linear map with one interior breakpoint moved off the diagonal serves when I is bounded, and the dilation $x \mapsto a + 2(x - a)$ about its finite endpoint a when it is not. Then $g(s) = s$, so $\tau(g(s)) = \tau(s) \neq g(\tau(s))$, contradicting equivariance. Hence $\tau(s) \in \{s_{(1)}, \ldots, s_{(n)}, +\infty, -\infty\}$. For two pools s, s' with distinct entries, the piecewise-linear interpolation of $s_{(j)} \mapsto s'_{(j)}$, extended linearly with slope one beyond the extreme values, is a strictly increasing bijection g of $\mathbb{R}$; $g(s)$ has the same entries as s', so by symmetry $\tau(s') = \tau(g(s)) = g(\tau(s))$. If $\tau(s) = s_{(k)}$ then $\tau(s') = s'_{(k)}$, and if $\tau(s) = \pm\infty$ then $\tau(s') = \pm\infty$. The index, or the infinite value, is therefore the same for every pool. □

Proof of Theorem 2. Reduction. Let F be the common continuous distri-

bution and put $u_i = F(s_i)$, $u'_i = F(s'_i)$. These are uniform on $(0,1)$ with the same independence structure, and almost surely $s'_i > s_{(k)}$ if and only if $u'_i > u_{(k)}$, because a disagreement requires two sample values in a closed interval on which F is constant, an event of probability zero. We may therefore take F uniform.

Upper bound. Condition on the pool. For each i the pool and u'_i are independent, so $P(u'_i > u_{(k)} \mid \text{pool}) = 1 - u_{(k)} =: q$. By the union bound and $P \le 1$, $P(\max_i u'_i > u_{(k)} \mid \text{pool}) \le \min\{1, mq\}$, whatever the dependence among the u'_i. Taking expectations gives $\mathbb{E}[\min\{1, mB\}]$ with $B = 1 - u_{(k)} \sim \text{Beta}(n+1-k, k)$.

Attainment. Let the pool be independent uniforms, let V be uniform and independent of the pool, and put $u'_i = \text{frac}\big(V + (i-1)/m\big)$. Each u'_i is uniform and a function of V alone, so the pool and u'_i are $n+1$ independent uniforms and the law lies in $\mathcal{Q}_{n,m}$. Given the pool, $u'_i > 1 - q$ if and only if V lies in an arc of length q of the circle $\mathbb{R}/\mathbb{Z}$ whose right endpoint is $1 - (i-1)/m$. The m right endpoints are equally spaced at distance $1/m$. If $q \le 1/m$ the arcs are pairwise disjoint and their union has measure mq; if $q > 1/m$ consecutive arcs overlap and the union is the whole circle. Hence $P(\max_i u'_i > u_{(k)} \mid \text{pool}) = \min\{1, mq\}$, and the upper bound holds with equality.

The fence's threshold. With $k = k^*$, $B \sim \text{Beta}(\rho, n+1-\rho)$ and $\mathbb{E}[B] = \rho/(n+1)$. Since $\min\{1, mB\} = mB - (mB-1)_+$, the worst case is $m\rho/(n+1) - \mathbb{E}[(mB-1)_+]$, and $m\rho/(n+1) \le \alpha$ by Proposition 5(b). Because $B \le 1$, $(mB-1)_+ \le (m-1)\,\mathbf{1}\{B > 1/m\}$. The variable B is distributed as the ρ-th smallest of n independent uniforms, so $B > 1/m$ exactly when fewer

than ρ of them fall at or below $1/m$: $P(B > 1/m) = P(X \le \rho - 1)$ with $X \sim \mathrm{Bin}(n, 1/m)$. Put $\mu = n/m$. Then $\rho - 1 \le (n+1)\alpha/m - 1 < \alpha\mu$, and the Chernoff bound $P(X \le (1-\delta)\mu) \le \exp(-\delta^2\mu/2)$ with $\delta = 1 - \alpha$ gives $P(B > 1/m) \le \exp(-(1-\alpha)^2 n/(2m))$. □

Proof of Corollary 2. (a) For $m = 1$, $(B-1)_+ = 0$, so the worst case is $\rho/(n+1)$, which is α when $(n+1)\alpha$ is an integer. For general m, $m\rho/(n+1) \ge \alpha - m/(n+1)$ and the correction term of Theorem 2 tends to zero, so the worst case tends to α and exceeds any $\alpha' < \alpha$ for large n.

(b) For $m = 1$ the worst case of rank k is $\mathbb{E}[B] = (n+1-k)/(n+1)$, and $k < k^*$ gives $n+1-k \ge \rho+1 > (n+1)\alpha$. For a fixed $\gamma \in (\alpha/m, 1)$ let $k_\gamma = \lceil (n+1)(1-\gamma) \rceil$, which lies in $\{1, \dots, n\}$ for large n, and let $B_\gamma \sim \mathrm{Beta}(n + 1 - k_\gamma, k_\gamma)$. Its mean tends to γ and its variance is at most $1/(4(n+2))$, so $B_\gamma \to \gamma$ in probability; as $x \mapsto \min\{1, mx\}$ is bounded and continuous, the worst case tends to $\min\{1, m\gamma\}$, which exceeds α because $m\gamma > \alpha$ and $\alpha < 1$. The level α/m itself is valid for every n by Theorem 1.

(c) Let $m > \alpha(n+1)$ and $k \le n$. On the same pool $u_{(k)} \le u_{(n)}$, so $1 - u_{(k)} \ge 1 - u_{(n)}$ and the worst case of rank k is at least that of rank n, for which $B \sim \mathrm{Beta}(1, n)$ and $P(B > b) = (1-b)^n$. Then

$$\begin{aligned}\mathbb{E}[\min\{1, mB\}] &= \int_0^1 P(mB > u)\, du = \int_0^1 \left(1 - \frac{u}{m}\right)^n du \\ &= \frac{m}{n+1}\left(1 - \left(1 - \frac{1}{m}\right)^{n+1}\right).\end{aligned}$$

Put $x = m/(n+1)$. Since $(1-1/m)^{n+1} \le \exp(-(n+1)/m) = \exp(-1/x)$, this is at least $h(x) = x(1 - \exp(-1/x))$. The function h is continuous and strictly increasing on $(0, \infty)$, because $h'(x) = 1 - \exp(-1/x)(1 + 1/x) > 0$; it tends to 1 as $x \to \infty$; and $h(\alpha) < \alpha$. Hence there is a unique $x_\alpha > \alpha$ with

$h(x_\alpha) = \alpha$. With $\delta_\alpha = x_\alpha - \alpha$, every $x > \alpha + \delta_\alpha$ gives a worst case above α. Solving numerically, $\delta_{0.05} = 1.03 \times 10^{-10}$ and $\delta_{0.01} = 1.7 \times 10^{-18}$. For $\alpha = 1/20$, $m > (n+1)/20$ implies $20m \geq n + 2$, so $x - \alpha \geq 1/(20(n+1))$, which exceeds $\delta_{0.05}$ whenever $n+1 < 4.8 \times 10^8$. The threshold $-\infty$ suppresses every request. Under the stated condition, every scale-free threshold other than $+\infty$ is therefore invalid on $\mathcal{Q}_{n,m}$. □

Proof of Corollary 1. (a) Fix a class c and condition on it. The proof of Theorem 1 uses only that each distinct comparison score of the request is exchangeable with the pool on which the threshold is computed; under A6 within class c this holds for the class-c pool of size n_c, and the argument goes through with n_c in place of n. (b) The first statement is Proposition 5(a) with n_c in place of n. Every class is certifiable iff each $n_c \geq \lceil m/\alpha \rceil - 1$, i.e. iff $\min_c n_c$ is; summing over the K classes gives $n \geq K(\lceil m/\alpha \rceil - 1)$. Since $\min_c n_c \leq n/K$ and $x \mapsto \lfloor \alpha(x+1) \rfloor$ is non-decreasing, $\min_c \lfloor \alpha(n_c+1) \rfloor \leq \lfloor \alpha(n/K+1) \rfloor$. (c) Let the request be of class c. If $S^* > \max_{c'} t_{c'}(m)$ then $S^* > t_c(m)$, so the suppression event under the maximum is contained in the suppression event under $t_c(m)$, whose probability given class c is at most α by (a). If some class is not certifiable, $\max_{c'} t_{c'}(m) = +\infty$. □

Proof of Proposition 6. Let a threshold rule suppress iff the score exceeds τ. Zero wrong suppression requires that F_{diff} place no mass above τ. Zero silent escape without a band requires that F_{same} place no mass at or below τ. Together these require that every score with positive probability under F_{diff} be at most τ and every score with positive probability under F_{same} exceed τ, contradicting overlap. With a band $(t_{\text{lo}}, t_{\text{hi}}]$, silent escape can be removed by widening the band over the overlap, but the duplicates in the overlap are

then flagged rather than suppressed, so deduplication falls by the overlap mass; the error is relocated, not removed. A decision that does not consult the score is not subject to this argument. □

Proof of Proposition 7. Consider the first execution for *op* to reach the insert. By A3 the insert either creates the commit row or observes an existing one. Any later execution computing *op*—in the same process or a fresh one—computes the same identifier by A1 and A2, finds the committed row, and returns the recorded response on the certifier branch without dispatching. By A7 no second execution is inside the interval between its lookup and its insert while the first is, so the dispatch branch is entered at most once before a commit row exists, and by A4 that branch dispatches the effect once and commits only on success. If the process dies after the dispatch and before the insert, no commit row exists, the intent remains PENDING, and a later execution finds no row and may dispatch again; this is the stated exception. □

Proof of Proposition 8. Case A. If the fence suppresses, nothing is dispatched. If it executes and the process dies inside the commit window, a later execution re-dispatches the same act, possibly paraphrased. By paraphrase invariance both dispatches carry the same key, so by A8 the provider applies at most one effect; by completeness no distinct act carries that key, so no legitimate act is absorbed. Outside the window, Proposition 7 applies. If the schema satisfies A5″, σ_t is both invariant and complete, so $\pi_t = \sigma_t$ qualifies. *Case B.* Without A5″ there exist $x \not\approx_t x'$ with $\sigma_t(x) = \sigma_t(x')$; presented to a provider obeying A8 under key σ_t, the second is absorbed although it is a distinct act, which is a wrong suppression invisible to the

local log. The canonical identifier is admissible by Proposition 4 but changes under paraphrase, because κ_t normalises free text without interpreting it, so a paraphrased retry presents a new key and is applied. □

Proof of Proposition 9. If $x \approx_t x'$ but the fence computes different identifiers, the log's uniqueness constraint does not fire and a provider keyed on those identifiers receives two keys; under A8 it is obliged to apply both. If $x \not\approx_t x'$ but they receive the same identifier, both the log and the provider absorb the second, and both behave as specified. In either case the failure is attributable to identity recognition alone. Conversely, under Case A of Proposition 8 the provider applies at most one effect per act whatever the local fence serialised, so A7 is not needed for at-most-once application. □

References

[1] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, Y. Cao, ReAct: Synergizing Reasoning and Acting in Language Models, in: International Conference on Learning Representations (ICLR), 2023, arXiv:2210.03629.

[2] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, L. Zettlemoyer, N. Cancedda, T. Scialom, Toolformer: Language Models Can Teach Themselves to Use Tools, in: Advances in Neural Information Processing Systems 36 (NeurIPS), 2023, arXiv:2302.04761.

[3] S. Yao, N. Shinn, P. Razavi, K. Narasimhan, τ-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains, preprint, arXiv:2406.12045 (2024). doi:10.48550/arXiv.2406.12045.

[4] A. D. Birrell, B. J. Nelson, Implementing Remote Procedure Calls, ACM Transactions on Computer Systems 2 (1) (1984) 39–59. doi:10.1145/2080.357392.

[5] M. Kleppmann, Designing Data-Intensive Applications, O'Reilly Media, 2017.

[6] Stripe, Idempotent Requests, API documentation, https://docs.stripe.com/api/idempotent_requests, accessed 14 September 2026 (2026).

[7] Temporal Technologies, Temporal Documentation: Durable Execution, https://docs.temporal.io/, accessed 14 September 2026 (2026).

[8] Restate, Restate Documentation, https://docs.restate.dev/, accessed 14 September 2026 (2026).

[9] Y. Zheng, Y. Yang, W. Zhang, A. Quinn, ACRFence: Preventing Semantic Rollback Attacks in Agent Checkpoint-Restore, preprint, arXiv:2603.20625 (2026). doi:10.48550/arXiv.2603.20625.

[10] J. Xu, L. Fan, Z. Wang, X. Li, H. Liu, Beyond Single-Use Tokens: Durable Authorization State for Replay-Resistant LLM Agent Actions, preprint, arXiv:2608.01710 (2026). doi:10.48550/arXiv.2608.01710.

[11] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, I. Stoica, Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena, in: Advances in Neural Information Processing Systems 36 (NeurIPS), Datasets and Benchmarks Track, 2023, arXiv:2306.05685.

[12] P. Wang, L. Li, L. Chen, Z. Cai, D. Zhu, B. Lin, Y. Cao, L. Kong, Q. Liu, T. Liu, Z. Sui, Large Language Models are not Fair Evaluators, in: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, Bangkok, Thailand, 2024, pp. 9440–9450. doi:10.18653/v1/2024.acl-long.511.

[13] J. Ye, Y. Wang, Y. Huang, D. Chen, Q. Zhang, N. Moniz, T. Gao, W. Geyer, C. Huang, P.-Y. Chen, N. V. Chawla, X. Zhang, Justice or Prejudice? Quantifying Biases in LLM-as-a-Judge, preprint, arXiv:2410.02736 (2024). doi:10.48550/arXiv.2410.02736.

[14] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, M. Fritz, Not What You've Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection, in: Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security (AISec), 2023, pp. 79–90. doi:10.1145/3605764.3623985.

[15] Y. Hwang, D. Lee, T. Kang, Y. Kim, K. Jung, Can You Trick the Grader? Adversarial Persuasion of LLM Judges, preprint, arXiv:2508.07805 (2025). doi:10.48550/arXiv.2508.07805.

[16] P. Carbone, S. Ewen, G. Fóra, S. Haridi, S. Richter, K. Tzoumas, State Management in Apache Flink: Consistent Stateful Distributed Stream Processing, Proceedings of the VLDB Endowment 10 (12) (2017) 1718–1729. doi:10.14778/3137765.3137777.

[17] J. Gray, A. Reuter, Transaction Processing: Concepts and Techniques, Morgan Kaufmann, 1993.

[18] H. Garcia-Molina, K. Salem, Sagas, in: Proceedings of the 1987 ACM SIGMOD International Conference on Management of Data, 1987, pp. 249–259. doi:10.1145/38713.38742.

[19] S. Khan, Resume Means Resume: A Machine-Checked Conformance Contract for Checkpoint, Interrupt, and Resume Semantics in Workflow Persistence Layers, preprint, arXiv:2608.03836 (2026). doi:10.48550/arXiv.2608.03836.

[20] B. Mohammadi, N. Potamitis, L. Klein, A. Arora, L. Bindschaedler, Atomix: Timely, Transactional Tool Use for Reliable Agentic Workflows, preprint, arXiv:2602.14849 (2026). doi:10.48550/arXiv.2602.14849.

[21] H. Lyu, D. Zhang, M. Wu, X. Wei, H. Chen, CoAgent: Concurrency Control for Multi-Agent Systems, preprint, arXiv:2606.15376 (2026). doi:10.48550/arXiv.2606.15376.

[22] M. Balakrishnan, A. Bharambe, D. Testuggine, D. Geraghty, D. Mao, V. Venkat, I. Mironov, R. Baradi, G. Aiyer, V. Dudin, LogAct: Enabling Agentic Reliability via Shared Logs, preprint, arXiv:2604.07988 (2026). doi:10.48550/arXiv.2604.07988.

[23] B. Mozafari, BEGIN AI TRANSACTION: Semantic Isolation for Durable AI Workflows, preprint, arXiv:2608.05412 (2026). doi:10.48550/arXiv.2608.05412.

[24] I. K. Mansoor, A. Phadke, P. Rana, Verified Tool Calls Improve LLM Agent Reliability Under Non-Atomic Failures, preprint, arXiv:2608.02645 (2026). doi:10.48550/arXiv.2608.02645.

[25] Z. Yin, Safety Invariants for Agents Orchestrating Irreversible State Transitions: A Four-Dimensional Formalism Evaluated on Public Ledgers, preprint, arXiv:2608.00783 (2026). doi:10.48550/arXiv.2608.00783.

[26] A. K. Elmagarmid, P. G. Ipeirotis, V. S. Verykios, Duplicate Record Detection: A Survey, IEEE Transactions on Knowledge and Data Engineering 19 (1) (2007) 1–16. doi:10.1109/TKDE.2007.250581.

[27] D. Cer, M. Diab, E. Agirre, I. Lopez-Gazpio, L. Specia, SemEval-2017 Task 1: Semantic Textual Similarity Multilingual and Crosslingual Focused Evaluation, in: Proceedings of the 11th International Workshop on Semantic Evaluation (SemEval-2017), 2017, pp. 1–14. doi:10.18653/v1/S17-2001.

[28] X. Fang, K. Tong, J. Liu, T. Ning, Y. He, SemHash-LLM: A Multi-Granularity Semantic Hashing Framework for Document Deduplication, preprint, arXiv:2607.01601 (2026). doi:10.48550/arXiv.2607.01601.

[29] W. Gill, M. Elidrisi, P. Kalapatapu, A. Ahmed, A. Anwar, M. A. Gulzar, MeanCache: User-Centric Semantic Caching for LLM Web Services, in: Proceedings of the IEEE International Parallel and Distributed Processing Symposium (IPDPS), 2025, arXiv:2403.02694. doi:10.1109/IPDPS64566.2025.00117.

[30] G. H. Chen, S. Chen, Z. Liu, F. Jiang, B. Wang, Humans or LLMs as the Judge? A Study on Judgement Biases, preprint, arXiv:2402.10669 (2024). doi:10.48550/arXiv.2402.10669.

[31] D. Li, R. Sun, Y. Huang, M. Zhong, B. Jiang, J. Han, X. Zhang, W. Wang, H. Liu, Preference Leakage: A Contamination Problem in LLM-as-a-judge, preprint, arXiv:2502.01534 (2025). doi:10.48550/arXiv.2502.01534.

[32] T. Yuan, Z. He, L. Dong, Y. Wang, R. Zhao, T. Xia, L. Xu, B. Zhou, F. Li, Z. Zhang, R. Wang, G. Liu, R-Judge: Benchmarking Safety Risk Awareness for LLM Agents, in: Findings of the Association for Computational Linguistics: EMNLP 2024, 2024, arXiv:2401.10019.

[33] Y. Ruan, H. Dong, A. Wang, S. Pitis, Y. Zhou, J. Ba, Y. Dubois, C. J. Maddison, T. Hashimoto, Identifying the Risks of LM Agents with an LM-Emulated Sandbox, in: International Conference on Learning Representations (ICLR), 2024, arXiv:2309.15817.

[34] E. Debenedetti, J. Zhang, M. Balunović, L. Beurer-Kellner, M. Fischer, F. Tramèr, AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents, in: Advances in Neural Information Processing Systems 37 (NeurIPS), Datasets and Benchmarks Track, 2024, arXiv:2406.13352.

[35] D. M. Zuvic, Capability Gates Are Not Authorization: Confused-Deputy Failures in LLM Agent Frameworks, preprint, arXiv:2606.28679 (2026). doi:10.48550/arXiv.2606.28679.

[36] L. Beurer-Kellner, B. Buesser, A.-M. Creţu, E. Debenedetti, D. Dobos, D. Fabian, M. Fischer, D. Froelicher, K. Grosse, D. Naeff, E. Ozoani, A. Paverd, F. Tramèr, V. Volhejn, Design Patterns for Securing LLM Agents against Prompt Injections, preprint, arXiv:2506.08837 (2025). doi:10.48550/arXiv.2506.08837.

[37] V. Vovk, A. Gammerman, G. Shafer, Algorithmic Learning in a Random World, Springer, New York, 2005.

[38] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, L. Wasserman, Distribution-Free Predictive Inference for Regression, Journal of the American Statistical Association 113 (523) (2018) 1094–1111. doi:10.1080/01621459.2017.1307116.

[39] A. N. Angelopoulos, S. Bates, Conformal Prediction: A Gentle Introduction, Foundations and Trends in Machine Learning 16 (4) (2023) 494–591. doi:10.1561/2200000101.

[40] V. Quach, A. Fisch, T. Schuster, A. Yala, J. H. Sohn, T. S. Jaakkola, R. Barzilay, Conformal Language Modeling, in: International Conference on Learning Representations (ICLR), 2024, arXiv:2306.10193.

[41] C. Mohri, T. Hashimoto, Language Models with Conformal Factuality Guarantees, in: Proceedings of the 41st International Conference on Machine Learning (ICML), Vol. 235 of Proceedings of Machine Learning Research, PMLR, 2024, pp. 36029–36047.
URL https://proceedings.mlr.press/v235/mohri24a.html

[42] A. Z. Ren, A. Dixit, A. Bodrova, S. Singh, S. Tu, N. Brown, P. Xu, L. Takayama, F. Xia, J. Varley, Z. Xu, D. Sadigh, A. Zeng, A. Majumdar, Robots That Ask For Help: Uncertainty Alignment for Large Language Model Planners, in: Proceedings of the 7th Conference on Robot Learning (CoRL), Vol. 229 of PMLR, 2023, arXiv:2307.01928.

[43] H. Sheng, X. Liu, H. He, J. Zhao, J. Kang, Analyzing Uncertainty of LLM-as-a-Judge: Interval Evaluations with Conformal Prediction, preprint, arXiv:2509.18658 (2025). doi:10.48550/arXiv.2509.18658.

[44] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, T. Schuster, Conformal Risk Control, in: International Conference on Learning Representations (ICLR), 2024, arXiv:2208.02814.

[45] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei, Learn then Test: Calibrating Predictive Algorithms to Achieve Risk Control, preprint, arXiv:2110.01052 (2021). doi:10.48550/arXiv.2110.01052.

[46] S. Bates, E. Candès, L. Lei, Y. Romano, M. Sesia, Testing for Outliers with Conformal p-values, The Annals of Statistics 51 (1) (2023). doi:10.1214/22-AOS2244.

[47] D. Ulmer, C. Zerva, A. F. T. Martins, Non-Exchangeable Conformal Language Generation with Nearest Neighbors, preprint, arXiv:2402.00707 (2024). doi:10.48550/arXiv.2402.00707.

[48] Z. Lin, Y. Li, N. Sarna, Y. Gao, M. von Gablenz, Domain-Shift-Aware Conformal Prediction for Large Language Models, preprint, arXiv:2510.05566 (2025). doi:10.48550/arXiv.2510.05566.

[49] M. Tursunbadalov, M. Tursunbadalov, A Quiet Failure in Calibrated Virtual Screening: Marginal Conformal Prediction Under-Covers the Minority Class, and a Class-Conditional Fix Recovers It, preprint, arXiv:2607.06605 (2026). doi:10.48550/arXiv.2607.06605.

[50] F. Pezoa, J. L. Reutter, F. Suarez, M. Ugarte, D. Vrgoč, Foundations of JSON Schema, in: Proceedings of the 25th International Conference on World Wide Web (WWW), 2016, pp. 263–273. doi:10.1145/2872427.2883029.

[51] P. Buneman, S. Khanna, W.-C. Tan, Why and Where: A Characterization of Data Provenance, in: Database Theory – ICDT 2001, Vol. 1973 of Lecture Notes in Computer Science, Springer, 2001, pp. 316–330. doi:10.1007/3-540-44503-X_20.

[52] M. Fréchet, Généralisation du théorème des probabilités totales, Fundamenta Mathematicae 25 (1935) 379–387. doi:10.4064/fm-25-1-379-387.

[53] L. Rüschendorf, Random Variables with Maximum Sums, Advances in Applied Probability 14 (3) (1982) 623–632. doi:10.2307/1426677.

[54] J. Galambos, I. Simonelli, Bonferroni-type Inequalities with Applications, Springer, New York, 1996.

[55] D. Hunter, An Upper Bound for the Probability of a Union, Journal of Applied Probability 13 (3) (1976) 597–603. doi:10.2307/3212481.

[56] K. J. Worsley, An Improved Bonferroni Inequality and Applications, Biometrika 69 (2) (1982) 297–302. doi:10.1093/biomet/69.2.297.

[57] V. Vovk, R. Wang, Combining p-values via Averaging, Biometrika 107 (4) (2020) 791–808. doi:10.1093/biomet/asaa027.

[58] V. Vovk, B. Wang, R. Wang, Admissible Ways of Merging p-values under Arbitrary Dependence, The Annals of Statistics 50 (1) (2022). doi:10.1214/21-AOS2109.

[59] S. Ovchinnikov, Invariant Functions on Simple Orders, Order 14 (4) (1997) 365–371. doi:10.1023/A:1006054508202.

[60] J.-L. Marichal, On Order Invariant Synthesizing Functions, Journal of Mathematical Psychology 46 (6) (2002) 661–676. doi:10.1006/jmps.2002.1418.

[61] E. B. Wilson, Probable Inference, the Law of Succession, and Statistical Inference, Journal of the American Statistical Association 22 (158) (1927) 209–212. doi:10.1080/01621459.1927.10502953.

[62] Q. McNemar, Note on the Sampling Error of the Difference Between Correlated Proportions or Percentages, Psychometrika 12 (2) (1947) 153–157. doi:10.1007/BF02295996.

[63] S. Holm, A Simple Sequentially Rejective Multiple Test Procedure, Scandinavian Journal of Statistics 6 (2) (1979) 65–70.

[64] K. Krippendorff, Reliability in Content Analysis: Some Common Misconceptions and Recommendations, Human Communication Research 30 (3) (2004) 411–433. doi:10.1111/j.1468-2958.2004.tb00738.x.

[65] J. L. Fleiss, Measuring Nominal Scale Agreement Among Many Raters, Psychological Bulletin 76 (5) (1971) 378–382. doi:10.1037/h0031619.

[66] J. Cohen, A Coefficient of Agreement for Nominal Scales, Educational and Psychological Measurement 20 (1) (1960) 37–46. doi:10.1177/001316446002000104.

[67] E. Turan, Oversight Has a Capacity: Calibrating Agent Guards to a Subjective, Fatiguing Human, preprint, arXiv:2606.08919 (2026). doi:10.48550/arXiv.2606.08919.